

Review article: Hepatic steatosis and its associations with acute and chronic liver diseases

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Summary

Background: Hepatic steatosis is a common finding in liver histopathology and the hallmark of metabolic dysfunction-associated steatotic liver disease (MASLD), formerly known as non-alcoholic fatty liver disease (NAFLD), whose global prevalence is rising.

Aims: To review the histopathology of hepatic steatosis and its mechanisms of development and to identify common and rare disease associations.

Methods: We reviewed literature on the basic science of lipid droplet (LD) biology and clinical research on acute and chronic liver diseases associated with hepatic steatosis using the PubMed database.

Results: A variety of genetic and environmental factors contribute to the development of chronic hepatic steatosis or steatotic liver disease, which typically appears macrovesicular. Microvesicular steatosis is associated with acute mitochondrial dysfunction and liver failure. Fat metabolic processes in hepatocytes whose dysregulation leads to the development of steatosis include secretion of lipoprotein particles, uptake of remnant lipoprotein particles or free fatty acids from blood, de novo lipogenesis, oxidation of fatty acids, lipolysis and lipophagy. Hepatic insulin resistance is a key feature of MASLD. Seipin is a polyfunctional protein that facilitates LD biogenesis. Assembly of hepatitis C virus takes place on LD surfaces. LDs make important, functional contact with the endoplasmic reticulum and other organelles.

Conclusions: Diverse liver pathologies are associated with hepatic steatosis, with MASLD being the most important contributor. The biogenesis and dynamics of LDs in hepatocytes are complex and warrant further investigation. Organellar interfaces permit co-regulation of lipid metabolism to match generation of potentially toxic lipid species with their LD depot storage.

1 | INTRODUCTION

Hepatic steatosis is a common finding in liver histology. While persistent hepatic steatosis does not invariably lead to progressive liver disease, it can elicit inflammation, developing into steatohepatitis, particularly in people with components of the metabolic syndrome, namely central obesity, hypertension, hypertriglyceridaemia, low high-density lipoprotein levels and/or high fasting blood glucose; genetic risk factors; or exposure to toxins like alcohol. Liver fibrosis, cirrhosis and hepatocellular carcinoma (HCC) can develop despite few or no symptoms of disease. Hepatic steatosis is the hallmark of metabolic dysfunction-associated steatotic liver disease (MASLD) but can be seen with many other acute and chronic liver diseases (CLDs). Hepatic steatosis can also accompany acute liver failure, which may be associated with intrinsic, or hepatocellular, injury.

Imaging findings consistent with hepatic steatosis are estimated to be present in 27.3%–28.8%^{1,2} of the non-institutionalised population of adults in the United States in data collected from 2017 to 2018 by the National Health and Nutrition Examination Survey. Predictors of hepatic steatosis in this population include diabetes or prediabetes, anthropometric measures such as waist-to-hip ratio and body mass index (BMI) and self-reported measures of physical inactivity, sedentary behaviour and unhealthy diet.² Hepatic steatosis is also an increasingly common finding in children.³ The prevalence of steatotic liver disease (SLD), its contribution to inflammatory, progressive steatohepatitis and the similarity of pathological findings among alcohol-related, metabolic dysfunction-associated and drug-induced SLD⁴ (pp. 746, 749) are challenging to pathologists.

The clinical definition of non-alcoholic fatty liver disease (NAFLD) was initially proposed to indicate a clinical and pathologic spectrum with hepatic steatosis as a common pathologic feature.^{5–8} Given the lack of connection with underlying mechanism of insulin resistance and potential stigma associated with term 'fatty', a multi-society process replaced the name NAFLD with MASLD.⁹ MASLD is hepatic steatosis accompanied by at least one cardiometabolic risk factor, defined using appropriate adult or paediatric criteria. MASLD may coexist with competing aetiologies of hepatic steatosis or CLD, an important difference from a NAFLD diagnosis, which is one of exclusion.⁹ The nomenclature consensus group⁹ additionally proposed that women who consume between 140 and 350 g of alcohol per week and men who consume between 210 and 420 g of alcohol per week, in the presence of hepatic steatosis and the cardiometabolic risk factors that define MASLD, receive a metabolic dysfunction and alcohol-associated steatotic liver disease (MetALD) diagnosis, distinct from alcohol-associated or alcohol-related liver disease (ALD). According to a recent study,¹⁰ the concordance between NAFLD and MASLD is excellent and therefore, both terminologies indicate generally the same liver disease. Globally, MASLD is the most common cause of CLD.^{11–13} Younossi et al.¹² report that NAFLD/MASLD is currently present in approximately 38% of the population and that the prevalence of this disease has increased greatly over the past 30 years.

The underlying pathologic mechanisms reflected by the development of hepatic steatosis can be different in different disease states. In MASLD, insulin resistance and lipid metabolism play an important role. An understanding of lipid metabolism in the liver may aid in distinguishing the aetiologies of hepatic steatosis and inform choice of further diagnostic studies. It is also important to understand how individual genetic variation influences the development of hepatic steatosis and can interact with environmental exposures, including hepatotoxic agents, to cause acute or CLD. The lipid droplet (LD) itself is a complex organelle whose structure, biogenesis and dynamics within hepatocytes are major subjects of research. In this review, we summarise key features of hepatic steatosis and recent developments in the field that are relevant to liver pathologists and researchers.

1.1 | Overview of liver lipid metabolism

Lipid metabolism in the liver involves several major pathways. These include the secretion of lipoproteins, uptake of lipoproteins, de novo lipogenesis, oxidation of fat and lipophagy. Impaired regulation of any of these pathways can result in hepatic steatosis. Secretion of lipoproteins, a unique function of the liver, takes place through the secretory pathway, with lipid loading occurring in apolipoprotein B-100 or ApoB-100-containing precursors within the lumen of the endoplasmic reticulum (ER).^{14,15} Apolipoprotein B transcripts are subject to RNA editing, resulting in the ApoB-48 isoform being produced in the intestines and incorporated into chylomicrons, whose remnants are taken up by the liver after passing through systemic circulation.^{14,15} Beta oxidation of fat is regulated by the activity of acetyl CoA carboxylase 2 within the mitochondrion, whose product, malonyl CoA, inhibits carnitine acyltransferase 1. Circulating fatty acids, mostly bound to albumin, can be taken up by hepatocytes using multiple protein transporters.¹⁶ Several master transcriptional regulators of lipid metabolism contribute to longer-term changes in gene expression, including members of the peroxisome proliferator-activated receptor (PPAR) family of transcription factors, PPAR gamma (PPARG) coactivator-1 alpha (PGC-1 α) and the transcription factor sterol regulatory element-binding protein 1c (SREBP-1c).¹⁷ The complexity of the regulatory frameworks makes their changes in liver pathology amenable to gene network pathway analysis.¹⁷

Insulin stimulates de novo lipogenesis and the formation of LDs¹⁸ and exerts anti-lipolytic effects.¹⁹ Andersson et al.¹⁸ identified ERK2 as a factor necessary for phospholipase D₁-mediated stimulation of LD formation in preadipocytes, possibly a result of ERK2 phosphorylation of dynein. Importantly, phospholipase D₁ was the major mediator of insulin-induced LD accumulation in the preadipocytes.¹⁸ Insulin also acts via the protein kinase B and phosphoinositol-3-kinase pathway to increase de novo lipogenesis under the regulation of the SREBP-1c transcription factor.²⁰ SREBP-1c is additionally inducible by liver X receptor agonism.²¹ The presence of LDs in a cell may affect its ability to respond to insulin—the SNARE protein SNAP23 was demonstrated by Boström et al.²² to be present on the

LD of preadipocytes and redistributed away from the plasma membrane in cardiomyocytes after oleic acid treatment, while SNAP23 overexpression specifically rescued insulin-stimulated glucose transporter type 4 (GLUT4) incorporation into the plasma membrane in the presence of exogenous oleic acid. It is believed that bioactive lipid species contribute to hepatic insulin resistance, including diacylglycerols (DAGs), which activate protein kinase C-epsilon, resulting in phosphorylation of the insulin receptor²³ and/or insulin receptor substrate-1 (IRS-1)²⁴ with downstream effects on insulin signalling, and ceramides, which activate protein kinase C-zeta to inhibit the Akt/protein kinase B pathway^{23,24} and stimulate activation of the fatty acid transport protein CD36.²³ Abdominal obesity and hypertension in combination is associated with decreased insulin-stimulated suppression of lipolysis in peripheral adipose tissue, and abdominal obesity itself may be associated with defects in whole-body lipid oxidation capacity.²⁵

2 | TYPES OF HEPATIC STEATOSIS AND THEIR DIAGNOSIS

2.1 | Differential diagnosis of diffuse and focal hepatic steatosis

Steatosis can be diffuse or a focal finding within the liver. Focal hypoxia may lead to development of an isolated steatotic lesion²⁶ (p. 333). Additionally, anomalous venous drainage of the pancreaticoduodenal vein, part of the parabiliary venous system, has been described in the context of a fatty lesion of Couinaud segment IV.²⁷ It is possible that focal fatty change is caused by exposure to local, relatively high concentrations of insulin supplied by these venous malformations.²⁷ Ectopic fat deposits of the coelom and mesenchymal tumours such as angiomyolipomas, myolipomas and lipomas have a similar microscopic appearance²⁶ (p. 333), although coelomic fat ectopias are described as being contained in a mesothelial lining.²⁸ Multifocal liver fat deposits have been described in human immunodeficiency virus (HIV) infection, where they may be confused with peliosis, granulomas, lymphoma or Kaposi's sarcoma.²⁹ HIV and its antiretroviral therapy can also give rise to generalised steatohepatitis, especially in the context of HIV-associated lipodystrophy syndrome⁴ (p. 749). A multifocal pattern of hepatic steatosis has also been seen in association with porphyria cutanea tarda and haemochromatosis.³⁰

Importantly, 'fatty changes' may also be detected in malignant liver tumours, such as HCC.³¹ A steatohepatic histologic variant of HCC was described by Salomao et al.³² among patients with hepatitis C-associated HCC, frequently with co-morbid obesity, type 2 diabetes or hypertension. It is unclear whether steatohepatic HCC occurs due to exposure of the tumour to steatogenic factors associated with an underlying metabolic syndrome or is due to de novo mutations.^{33,34} The expression of PAT family LD proteins, who will be discussed individually in the next section, is heterogeneously altered during hepatocarcinogenesis, with decreasing expression of

perilipin 1,^{31,35} potentially effected at a post-transcriptional level,³¹ and upregulation of adipophilin.^{31,35} Dynamic computed tomography (CT) and/or magnetic resonance imaging (MRI) data may help distinguish angiomyolipomas from these HCCs.³⁶

A finding of focal hepatic steatosis or diffuse steatosis with additional focal enhancement (hypersteatosis) or focal sparing may prompt consultation with a radiologist to determine need for further studies, such as contrast-enhanced ultrasound,³⁷ to exclude an alternative disease process. In summary, the differential diagnosis includes SLD, benign fatty tumours, ectopic coelomic fat or primary HCC. In patients with a prior history of malignancy, careful radiologic assessment can also help exclude metastasis. Focal sparing or steatosis is more likely to be benign if the lesion does not exhibit mass effect, is subcapsular, for example, near the porta hepatis or in the gallbladder fossa, and has expected shape or demarcation.³⁷ Diffuse hepatic steatosis, the more common appearance of SLD,³⁷ manifests as uniform hyperechogenicity on ultrasound, which can be compared to the hypoechogenic uninvolved spleen.

2.2 | Microscopic appearance of vesicular fat

Hepatic steatosis can be divided into two categories based on the predominant LD morphology: macrovesicular or microvesicular. Key features that distinguish hepatocytes with macrovesicular fat from those with microvesicular fat are the location and shape of the nucleus, which is displaced to the cell periphery by the large central LD of macrovesicular fat and often distorted, with increased elliptical eccentricity, but is centrally located and more round when surrounded by microvesicular LDs²⁶ (p. 331). Brunt³⁸ further distinguishes the latter category as 'small droplet' (Figure 1A) and 'true microvesicular' as true microvesicular fat droplets (Figure 1B) are often difficult to resolve by light microscopy, smaller than a micrometre in diameter³⁹ and cause enlargement of the affected hepatocytes,^{38,39} while hepatocytes containing small- or intermediate-sized fat droplets may accompany those with large fat droplets in MASLD, so-called 'mediovvesicular' steatosis²⁶ (p. 331). We refer to these small- and intermediate-sized fat droplets that do not displace the nucleus as small-droplet *macrovesicular* LDs and reserve the term microvesicular for the aforementioned 'true microvesicular' fat, but definitions used in the literature vary. The development of chronic macrovesicular hepatic steatosis may be viewed as a compensatory fat storage mechanism in the liver, but true microvesicular steatosis is generally caused by severe impairment in mitochondrial beta oxidation or gross mitochondrial injury, although exact mechanisms of the induced mitochondrial dysfunction may differ by agent⁴ (p. 746).

Most LDs are intracytoplasmic, but occasionally, intranuclear inclusions can be seen that may represent intranuclear LDs (Figure 2A). The mechanisms by which LDs form in the nucleus will be discussed in a subsequent section. Other apparent intranuclear inclusions may represent glycogen or glycogenated nuclei²⁶ (p. 364), where hepatocyte nuclei are occupied by glycogen-containing vacuoles that displace chromatin.⁴⁰ Schwertheim et al. have described

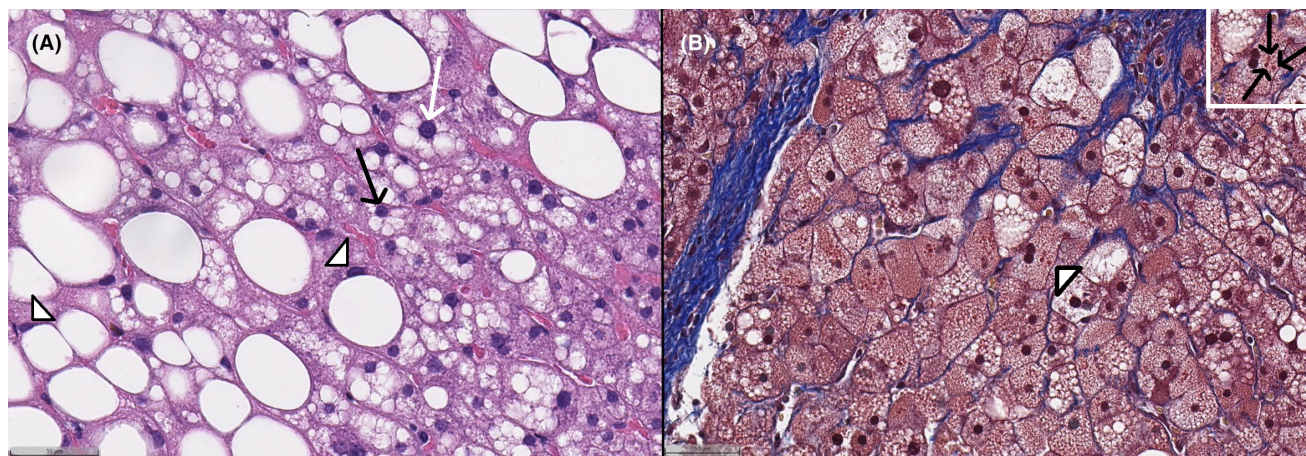


FIGURE 1 (A) Biopsy of metabolic dysfunction-associated steatotic liver disease showing combined large-droplet and small-droplet macrovesicular steatosis, with crescent-shaped, marginalised nuclei (arrowheads) in large-droplet and round nuclei (arrows) in small-droplet macrovesicular steatotic hepatocytes. Haematoxylin & eosin stain. (B) Biopsy of metabolic dysfunction-associated steatohepatitis showing microvesicular steatosis and megamitochondria (at arrowhead and in inset, individually indicated by arrows). Mallory's trichrome stain. Scale bars indicate 50 µm.

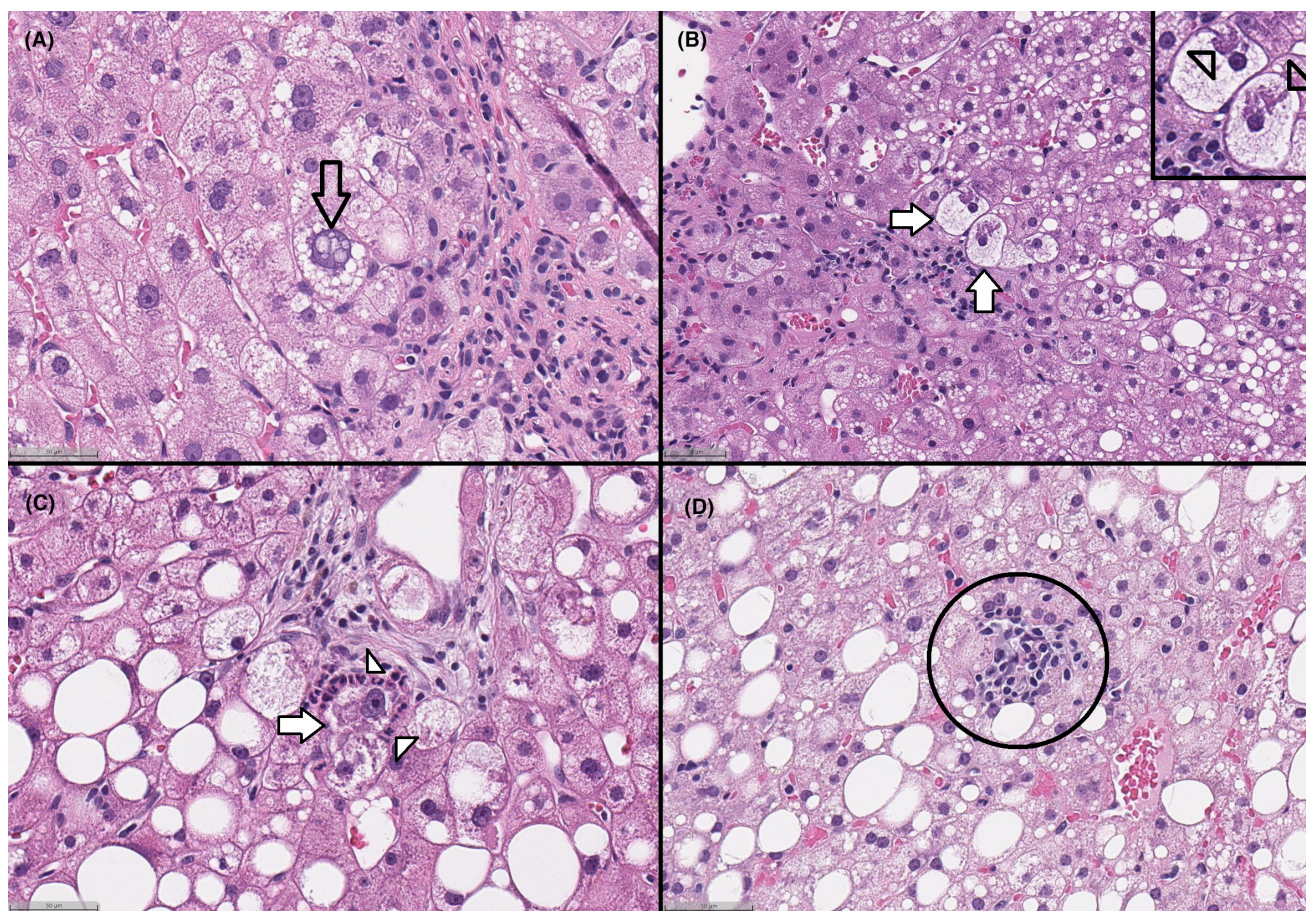


FIGURE 2 (A) Hepatocyte nucleus (arrow) with multiple intranuclear inclusions, possibly lipid droplets (LDs), in metabolic dysfunction-associated steatotic liver disease. (B) Ballooned hepatocytes (arrows) with intracytoplasmic, rope-like Mallory–Denk bodies (inset, arrowheads) and cytoplasmic clearing in metabolic dysfunction-associated steatohepatitis (MASH). (C) A ballooned hepatocyte (arrow) with neutrophil satellitosis (arrowheads) and an indistinct, likely degenerating plasma membrane in MASH. (D) A focus of lobular inflammation (circled) in MASH. Haematoxylin and eosin stains. Scale bars indicate 50 µm.

occupancy by ubiquitinated proteins, autophagy-related debris and heterolysosomes in a subset of intranuclear inclusions of MASLD hepatocytes.⁴⁰

The appearance of vesicular fat has several histologic mimics that can arise due to dysregulation of other cellular pathways. For example, in lysosomal acid lipase deficiency, vesicles arise from expansion of lysosomes. These can be distinguished by the presence of lysosomal markers such as cathepsin D and lysosome-associated membrane proteins 1 and 2 (LAMP1 and LAMP2), when examined by immunohistochemistry and by the presence of birefringence caused by cholesterol crystals in fresh tissue.^{41,42} Vacuolisation can also be induced by chemical agents, such as weakly basic compounds that are also lipophilic and can build up in acidic organelles, or during the degenerative changes induced by bacterial toxins or viral proteases, such as the 3C protease of hepatitis A virus.⁴³ Some of the markers that may be studied to distinguish these cases include Rab9, associated with late endosomes, Rab11, associated with vesicles that are recycled to the plasma membrane, and Rab5, associated with early endosomes.⁴³ Nor is hepatic steatosis solely caused by fatty transformation of hepatocytes—in Niemann–Pick type C disease, for example, hepatic steatosis occurs but is predominantly due to foamy-appearing Kupffer cells and infiltrating macrophages, although the lysosomal accumulation of unesterified cholesterol and other lipids progressively involves hepatocytes⁴⁴ (pp. 207–208).

2.3 | Pathologic workup

Hepatic steatosis may be seen in cases of isolated hepatic dysfunction, as a hepatic manifestation of the metabolic syndrome, in which hyperinsulinaemia and hyperglycaemia lead to increased de novo lipogenesis in the liver,⁴⁵ in viral infection or be a part of a toxidrome. Clinical correlation by the pathologist is important, as is a search for tissue inflammation, fibrosis or necrosis that may indicate acute or chronic steatohepatitis or, especially when necrosis is extensive, acute liver failure. Liver toxicity is a rare cause of hepatic steatosis and is often associated with cumulative exposure to the offending agent.⁴⁶ A history of parenteral nutrition,⁴⁷ which may induce essential fatty acid deficiency⁴⁸; pancreatic exocrine insufficiency with⁴⁹ or without⁵⁰ a history of extensive abdominal surgery; partial lipodystrophy or congenital total lipodystrophy⁴⁴ (p. 230); or a past medical history of childhood severe wasting, which may predispose to obesity-related SLD⁵¹ should be sought. Glucocorticoid therapy most frequently produces steatosis without steatohepatitis⁴ (p. 749).

The pattern of fat distribution within the microscopic architecture of the liver may provide a clue as to pathophysiology. Hepatocytes can be divided into three zones based on their position in their liver acinus and pattern of blood flow within the liver, with zone 1 closest to the oxygenated portal blood supply and zone 3 closest to the central vein. Classically, MASLD and ALD initially result in fatty transformation of zone 3 hepatocytes, extending to zone 2 in MASLD or to all zones in ALD as the transformation progresses²⁶ (pp. 342, 359–360). However, paediatric MASLD may have

steatosis restricted to periportal areas or be perivenular or azonal²⁶ (pp. 366–368). In ALD, chronic ethanol ingestion leads to induction of cytochrome P450 2E1²⁶ (p. 337), an important component of the microsomal pathway of ethanol metabolism that primarily localises to zone 3 but extends to the midzonal hepatocytes in active consumers of alcohol.^{52,53} Low oxygen tension developing in zone 3 of livers affected by ALD likely contributes to locally altered redox potential and steatosis development²⁶ (p. 340), which will be discussed later in more detail. Steatosis in chronic hepatitis C (CHC) is often azonal and generally mild⁵⁴ (pp. 434–435). Steatosis due to wasting in HIV/acquired immunodeficiency syndrome (AIDS) or use of glucocorticoids may begin adjacent to portal areas²⁶ (p. 331).

An important task for pathologists examining liver tissue with steatosis is to assess the presence or absence of steatohepatitis. Metabolic dysfunction-associated steatohepatitis (MASH) and alcohol-related steatohepatitis (ASH) are associated with hepatocytes undergoing ballooning degeneration, Mallory–Denk bodies (MDBs) and lobular inflammation. Ballooning hepatocytes may or may not contain visible fat droplets and can be recognised as being 1.5 or more times larger than non-steatotic, non-ballooning hepatocytes.⁵⁵ Their nuclei may be hyperchromatic and the cell membrane may be pronounced. MDBs may be present in a subset of ballooned hepatocytes²⁶ (pp. 333–334; **Figure 2B**). Cytoskeletal dysfunction is believed to be responsible for hepatocellular ballooning degeneration²⁶ (pp. 333–334), visible as clumping or rarefaction of the cytoplasm, but the fate of ballooned hepatocytes and the process of ballooning are incompletely understood. An extended hepatocyte ballooning score for assessment of NAFLD has recently been described as an aid in the diagnosis of borderline steatohepatitis cases⁵⁵; it incorporates a measure of ‘non-classic’ ballooning, which scores hepatocytes that, like classically ballooned hepatocytes, display cytoplasmic alteration but do not strictly meet criteria for enlargement and generally lack developed MDBs. Neutrophil satellitosis can occasionally be observed as a rim of polymorphonuclear cells around a ballooned hepatocyte²⁶ (p. 335; **Figure 2C**).

González et al.⁵⁶ found that MDBs were mostly distributed to zone 1 in amiodarone-induced liver injury, while they were preponderant in zone 3 or distributed azonally in ASH. As previously discussed, MDBs contain p62, which has a role in chaperone-mediated autophagy, as well as ubiquitin-modified and phosphorylated proteins²⁶ (pp. 334–335), suggesting decreased function of the proteasome. By electron microscopy, MDBs may be classified into three types depending on their pattern of organisation: those with parallel filaments, randomly arranged filaments or amorphous or granular structures²⁶ (pp. 334–335). Additionally, MDBs stain positive for cytokeratins 8 and 18, reflective of intermediate filament dysfunction²⁶ (pp. 334–335). Weakly organised lipogranulomas in addition to foci of mixed lobular inflammation (**Figure 2D**) may be observed in steatohepatitis. Fibrosis in steatohepatitis is typically perivenular and/or pericellular initially²⁶ (p. 336). Steatosis may become less prominent or even absent as necroinflammation and fibrosis progress in MASH²⁶ (p. 360).

2.4 | Diagnostic modalities

Hepatic steatosis has been variously defined as the occupancy of the liver by >5% fat by weight, >5% of hepatocytes by intracellular LDs or, in the context of CHC, >1% of hepatocytes by macrovesicular LDs.⁵⁷ Histologically, mild macrovesicular steatosis from 5% to 33% of hepatocytes or biopsy area is defined as grade 1, up to 66% for grade 2, while grade 3 indicates greater than 66% fat occupancy.^{57,58} Liver fat by weight-based criteria can be estimated by multiple imaging techniques, including ultrasound, magnetic resonance spectroscopy, MRI, MRI proton density fat fraction (MRI-PDFF)⁵⁸ and, though less preferred due to radiation exposure, CT.⁵⁹ MRI-PDFF has not been validated for assessment of microvesicular steatosis. It is, however, highly specific in detecting macrovesicular steatosis.^{58,60} Ultrasound is insensitive to microvesicular steatosis,⁶¹ but both ultrasound and MRI are suitable for detecting severe fatty infiltration of the liver.^{59,62,63} The data about the utility of imaging technologies such as the transient elastography-based controlled attenuation parameter for assessment of progression or regression of hepatic steatosis in clinical practice are not established.⁶⁴

Blood test indices for inferring the presence of hepatic steatosis have been developed. For example, the hepatic steatosis index, Framingham steatosis index and Dallas steatosis index use formulas that combine commonly measured blood tests, like aminotransferases, fasting glucose and/or serum triglycerides, with demographic and clinical variables to estimate likelihood of hepatic steatosis.⁶⁵ The fatty liver index incorporates measurement of waist circumference and gamma-glutamyl transferase,⁶⁶ and a derivative U.S. fatty liver index has been developed for use in the population of the United States.⁶⁷ These measures have moderate accuracy in predicting hepatic steatosis, with areas under the receiver-operating curve ranging from approximately 0.72–0.83.^{65,67,68} They may have utility in ruling in SLD in localities where diagnostic imaging studies are inaccessible. These indices are often used in public health research on SLD involving population-based, observational case-control studies.

Liver biopsy is generally reserved for cases where the aetiologies of SLD are unknown or the severity of CLD is not able to be determined in its absence, such as in the presence of prolonged unexplained elevations in transaminases or other liver chemistries or if there is suspicion of advanced hepatic fibrosis with equivocal non-invasive/less invasive test results.⁶⁴ Percutaneous liver biopsy is associated with risk of a major bleed that occurs in less than 2% of recipients, infection, sepsis and organ perforation.⁶⁹ Transvenous liver biopsies may be a safer alternative for patients with ascites or coagulopathy.⁶⁹ There is also risk of sampling error (missing a lesion with the biopsy needle) and error in interpretation or communication.

3 | BIOLOGY OF THE LD

3.1 | Composition

Biochemically, the LDs of hepatocytes contain neutral lipid species,⁷⁰ including cholesteryl esters (CEs) and triglycerides, although

the presence of other lipid constituents has not been excluded.⁷¹ The macrovesicular LD is lined by a phospholipid monolayer.⁷² Microvesicles have been reported to appear non-membrane bound in electron micrographs of cases of acute liver failure with hepatic steatosis.³⁹ It is speculated that they are instead lined by amphiphilic free fatty acids.⁷³ Microvesicular LDs will not be discussed further in this section.

Monotopic integral and peripheral membrane proteins are present at the surface of the LD⁷⁴ (Figure 3A). LD-associated proteins comprise a diverse group of 100–150 proteins,⁷⁴ whose relevance in protection against and development of hepatic steatosis is becoming steadily more appreciated. Trafficking of LD-associated proteins is not well understood⁷⁵; targeting of caveolin protein to LDs involves a hydrophobic domain and N-terminally adjacent basic residues,⁷⁶ while an amphipathic helix and angulation induced by proline residues may help 17-beta-hydroxysteroid dehydrogenase 13 segregate preferentially into the phospholipid monolayer of LDs.⁷⁷ The incorporation of amphipathic helices into LDs, where intramembrane domains likely provide surfactant activity, has been suggested by Chorlay and Thiam⁷⁸ to be dependent on the neutral lipid content of the LD.

Important proteins that localise to the LD membrane include the perilipins, also known as the PAT family after three of its five mammalian members: perilipin 1 (PLIN1), adipophilin or adipose differentiation-related protein (ADRP or PLIN2) and tail-interacting protein of 47kDa (TIP47 or PLIN3).⁷⁹ Perilipin and ADRP associate with CGI-58, also known as ABHD5, a co-activator of adipose triglyceride lipase (ATGL), on the LD membrane monolayer.⁸⁰ The gene encoding CGI-58, *ABHD5*, is mutated in neutral lipid storage disease with ichthyosis, causing accumulation of triglycerides and possibly impairment in phospholipid metabolism.^{81,82} CGI-58 is also a lysophosphatidic acid acyltransferase.⁸³ Perilipin A, the best-studied isoform of perilipin 1, and CGI-58 are both phosphorylated by protein kinase A (PKA) to facilitate dispersion of CGI-58 from the LD so that it can interact with ATGL⁸⁴ to promote lipolysis.⁸⁵ Mouse perilipin A can recruit hormone-sensitive lipase (HSL) to LDs after phosphorylation of three HSL N-terminal serine residues by PKA.⁸⁶ Factors contributing to localisation of PAT family proteins to the LD are not completely understood,⁸⁷ but LD localisation of PLIN1, PLIN2 and TIP47 appears to require 11-mer repeated motifs common to all PAT family members that are predicted to form amphipathic helices.⁸⁸ In regard to their expression in hepatic steatosis, Hashani et al.⁸⁹ demonstrated trends towards increasing liver protein levels of PLIN1, PLIN2, PLIN5 and CGI-58 that contrasted with a decrease in CAV2 caveolin protein as hepatic triglyceride content increased in human liver specimens, although the expression pattern of the proteins varied considerably between individual cases of hepatic steatosis.

3.2 | Biogenesis

Formation of CE-containing LDs is promoted by triglycerides,⁹⁰ and both CEs and triglycerides interact with an integral membrane

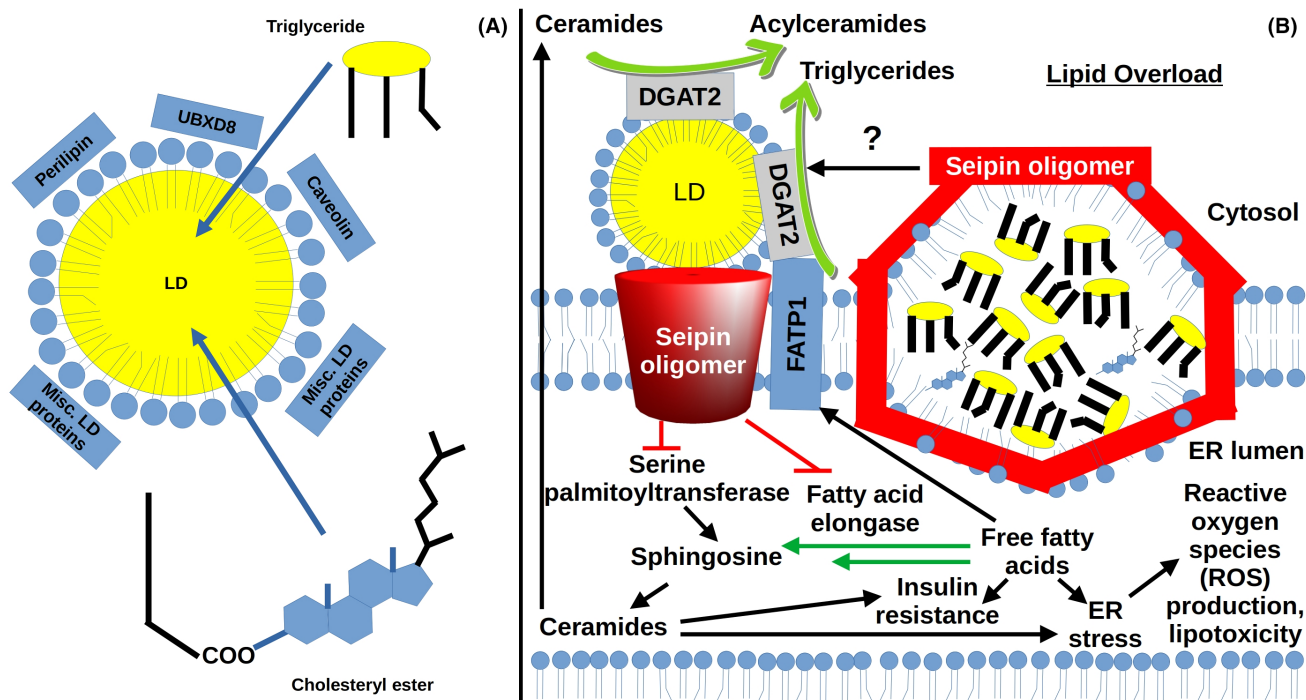


FIGURE 3 (A) A diagram of a lipid droplet (LD) showing neutral lipid species including cholesteryl esters (CEs) and triglycerides within a phospholipid monolayer. LD-associated proteins, which may be peripheral or monotopic integral,⁷⁴ are shown, including perilipin, ubiquitin-X domain adaptor 8 (UBXD8) and caveolin. (B) The roles of seipin, a protein involved in biogenesis of LDs, and LD-associated diacylglycerol acyltransferase 2 (DGAT2) enzyme in mitigating lipotoxicity under conditions of lipid overload at the ER-LD junction. Oligomers of the yeast orthologue of seipin can form a cage-like structure¹³⁹ and concentrate CEs and triglycerides within lens-shaped subdomains of the ER.⁷⁰ A predicted conformational change among transmembrane domains of seipin oligomers may facilitate LD budding¹³⁹ but has not been validated. Incorporation of diacylglycerol (DAG, not shown) into the growing blister within the ER membrane assists in expansion and curvature generation.⁹⁹ Yeast seipin additionally inhibits the activity of serine palmitoyltransferase and fatty acid elongase, which are involved in ceramide production.¹³⁷ Accumulation of ceramides¹⁴⁰ or free fatty acids^{122,127} is associated with ER stress and insulin resistance, which may lead to a positive feedback loop of increased peripheral lipolysis, further hepatic lipid overload and accumulation of reactive oxygen species (ROS). Conversion of the toxic lipid species ceramide into acylceramide for detoxification and sequestration among neutral lipids has been demonstrated in yeast¹⁴² and recently was shown to be dependent on LD-associated DGAT2 in lipid-overloaded colorectal carcinoma cells.¹⁴¹ Esterification of free fatty acids into triglycerides occurs at the ER-LD interface by interaction between ER-resident fatty acid transport protein 1 (FATP1) and LD-associated DGAT2.¹⁴⁴

protein of the ER, seipin, via their carboxyl ester linkages for enrichment and to nucleate LDs.⁷⁰ Loss-of-function mutations in the gene encoding seipin, *BSC12*, are a cause of congenital generalised lipodystrophy, or Bernadelli-Seip lipodystrophy, which features hepatic steatosis.⁹¹ LD nucleation is favoured by tubule morphology of the ER, also a site of seipin enrichment, while elongated sheets of ER rarely produce LDs and may produce larger LDs than tubular ER.⁹² Overexpression of seipin interactor Ldo45 results in mislocalisation of LD-resident proteins and morphologic defects in the LDs, resembling what is seen in a seipin knockdown phenotype.⁹³ Thus, seipin enables redistribution of LD-associated proteins from the ER to the LD and is required for normal LD biogenesis to occur.

Phospholipase A₂ activity is additionally required for LD production, and its inhibition under lipid load alters the structure of the ER in CHO-K1 cells.⁹⁴ Seipin can bind anionic phospholipids using an eight-stranded beta sandwich that lies within the ER lumen and resembles the sterol-binding domain of the Niemann Pick C2 protein,⁹⁵ suggesting a role for phospholipase substrates in LD biogenesis.⁹² DAGs, a product of phospholipase C catalysis, may be important in

LD biogenesis, as knockout of a phosphatidic acid phosphohydrolase orthologous to mammalian lipin⁹⁶ decreases LD production in yeast.⁹⁷ DAGs also recruit TIP47 to the LD,⁹⁸ and the enrichment of DAGs by seipin into the lentiliform ER membrane subdomain that serves as an LD precursor greatly aids in expansion of the triglyceride and DAG-containing blister.⁹⁹

Fundamental biophysical properties and processes that affect LD biogenesis include the Laplace pressure, which is inversely proportionate to LD radius, and Ostwald ripening, which favours diffusion-mediated growth of large LDs at the expense of smaller ones.^{92,100} Triglycerides can accumulate in ER membranes if they are not able to properly condense into LDs.⁹² According to Chung et al.,¹⁰¹ an association between seipin and lipid droplet assembly factor 1 (LDAF1) in the ER phospholipid bilayer facilitates LD formation when ER membrane concentrations of triglycerides are low, while redistribution of LDAF1 to the LD promotes budding from the ER, perhaps related to surfactant properties of the protein. Budding of the LD is also assisted by fat storage-inducing transmembrane (FIT) proteins.¹⁰² The interplay between relative cholesterol and triglyceride content

in intracellular membranes and LD formation is complex and is the subject of ongoing study.^{90,103}

3.3 | Role of lipophagy in LD catabolism

Lipophagy is an important contributor to the catabolism of hepatocellular LDs. In lysosomal acid lipase deficiency, including Wolman disease and cholesteryl ester storage disease, a deficiency of lysosomal acid lipase causes accumulation of fats in Kupffer cells and in hepatocytes in the form of small vacuoles that represent distended lysosomes, not true LDs²⁶ (p. 333).^{41,104} The expression of two types of lysosomal acid lipase is responsive to nutrient status in *Caenorhabditis elegans*.¹⁰⁵ Two possible routes for lipophagy include macroautophagy and microautophagy, which differ in that macroautophagy requires construction of an autophagosome via a double-membrane phagophore intermediate, while microautophagy, an important pathway for LD metabolism in yeast during nitrogen starvation¹⁰⁶ and in stationary phase,¹⁰⁷ involves direct contact and engulfment of LDs by the yeast vacuole, a structure analogous to the lysosomes of mammals.¹⁰⁶ Phosphorylation of perilipin 1 can increase the association of LDs with Rab7,¹⁰⁸ while Rab7 activation by starvation of hepatic cell lines promotes the colocalisation of multivesicular bodies, an intermediate in macroautophagy, with LDs, as well as that of lysosomes and LDs.¹⁰⁹ Dynamin 2 facilitates continued lipophagy by promoting scission of autolysosome tubules for autophagic lysosome reformation.¹¹⁰ Autophagic and proteasomal degradation of ApoB-100 proteins have also been localised to a peri-LD location in hepatocytes,¹¹¹ where lipidated ApoB-100 exists in crescents.¹¹² The mammalian target of rapamycin complex 1 (mTORC1) responds to changes in cell energy levels by negatively regulating autophagy and increasing lipogenesis via enhancement of SREBP activity.¹¹³ Lipophagy and mitochondrial beta oxidation have in common the protein ATG9A, a scramblase that aids in phagophore membrane expansion and fatty acid transfer to mitochondria from LDs.¹¹⁴ Interestingly, LDs also form under starvation conditions secondary to mTORC1-mediated autophagy, likely to accommodate the free fatty acids that are released when organelles are degraded through re-esterification.¹¹⁵

3.4 | Expansion and contraction of the cellular LD compartment buffers against lipotoxicity

The expansion and contraction of the hepatic LD compartment is a regulated process, in part to prevent lipotoxicity from free fatty acids. The LD-associated protein^{116,117} ubiquitin-X domain adaptor 8 (UBXD8) basally inhibits esterification of fatty acids to triglycerides, which can be relieved by provision of exogenous unsaturated fatty acids, simulating conditions of lipid overload.¹¹⁸ Cells with higher levels of fatty acid desaturases are resistant to lipotoxicity and incorporate more unsaturated and saturated fatty acids into their triglycerides,¹¹⁹ a finding that prompted Listenberger et al.¹¹⁹ to suggest that

unsaturated fatty acids may either increase the stability or solubility of LDs containing their triglyceride esters or serve as a preferred substrate for triglyceride synthesis. LDs have many contacts with mitochondria in hepatocytes, and understanding of these interactions and their mediators is under development.¹²⁰ Severe impairment of mitochondrial beta oxidation leads to microvesicular steatosis, in which excess fatty acids and dicarboxylic acid derivatives may further interfere with mitochondrial metabolism.^{73,121} Elevated levels of free fatty acids in serum can contribute to the development of insulin resistance, including decreased muscle glycogenesis and impaired glucose oxidation, possibly related to diminished glucose transport or its phosphorylation,¹²² and stimulate increases in pancreatic beta cell mass and insulin secretion that contribute to hyperinsulinaemia.^{123,124} The PATs or perilipins serve as important negative regulators of LD catabolism from their positions on the LD surface.^{125,126}

Key sensors of ER stress, IRE1 and PERK, are sensitive to the degree of fatty acid saturation in the ER membrane and activate the unfolded protein response in response to increasing saturation,¹²⁷ while LD formation is stimulated by the ER stress response¹²⁸ with unknown effects on the composition of ER membranes. Interestingly, the E3 ubiquitin ligase MARCH6, which functions in ER-associated protein degradation (ERAD),¹²⁹ was identified in a whole-genome CRISPR-CAS9 screen performed by Roberts et al.¹³⁰ as causing PLIN2 and LD accumulation in knockout cell lines, suggesting a role for ERAD in regulating cellular LD content. Oxidative stress and phospholipid or CE peroxidation appear to be involved in progression of steatosis to steatohepatitis^{131,132} and may also be a subject of regulation by LDs.¹³³

3.5 | The endoplasmic reticulum–lipid droplet interface regulates LD depot function

Many cellular LDs associate with the ER membrane in vivo,¹³⁴ and the ER–LD interface is important for regulation of lipid metabolism. Long-chain acyl-coenzyme A synthetase 3 (ASCL3) distributes to LDs as they form¹³⁵ and may have an important role in supplying fatty acids for triglyceride synthesis while the nascent LD is associated with the ER.¹³⁶ In yeast, the ER–LD interface is the site of inhibition of the ceramide-producing enzymes serine palmitoyltransferase and fatty acid elongase by the seipin orthologue, and expression of human seipin rescues sphingolipid production inhibitor-induced growth phenotypes in seipin knockout yeast.¹³⁷ Seipin proteins are targeted by polyunsaturated fatty acids and cyclopropane fatty acids to ER subdomains involved in LD development,¹³⁸ where the seipin oligomer forms cage-like structures.¹³⁹ Accumulation of ceramide can induce cellular insulin resistance by multiple mechanisms,¹⁴⁰ and its conversion to acylceramide by LD-associated diacylglycerol acyltransferase 2 (DGAT2)¹⁴¹ may help shunt the toxic species into neutral lipid storage.^{142,143} Furthermore, interaction between the ER protein fatty acid transport protein 1 (FATP1) and LD-associated DGAT2 facilitates LD growth,¹⁴⁴ sequestering free fatty acids. Thus, LDs, particularly when they are allowed to associate with the

ER, are important to the maintenance of cellular lipid homeostasis (Figure 3B). Emerging evidence suggests that the localisation of LDs is a regulated process that may influence LD growth and breakdown. In yeast, a subpopulation of LDs containing Pdr16, a phosphatidylinositol transfer protein, localises to the nucleus–vacuole junction in association with the seipin interactors Ldo45 and Ldo15.⁹³

3.6 | Structure and function of LD-associated proteins: Insights from HCV infection

Hepatitis C virus (HCV) exhibits dependence on hepatocellular LDs for replication and itself induces LD formation. Classically, chronic infection by genotype 3^{57,145,146} or 3a¹⁴⁷ is associated most strongly with hepatic steatosis, and viral load correlates with steatosis grade in the case of genotype 3.¹⁴⁸ The core protein of HCV includes, in its second domain (D2), two alpha helices joined by a hydrophobic loop containing proline residues at positions 138 and 143 analogous to the proline knot within oleosin plant LD proteins.^{149,150} Core protein localises to LDs in a D2-dependent manner and is important for viral assembly and release.^{71,149,150} Mutation of the two key proline residues to alanines completely abolishes human hepatoma cell production of infectious JFH1, an HCV genotype 2a derivative, corresponding to lack of LD coating by core at expected time points post-transfection and maintenance of a punctate distribution only seen at early time points after transfection by wild-type JFH1, which Boulant, Targett-Adams and McLauchlan speculate may represent core accumulation at ER-LD junctions.¹⁵¹ Non-structural proteins, such as NS5A, are recruited to LDs by core,^{152,153} and remain localised to the cytoplasmic face of the ER in cells replicating the JFH1 P138A/P143A core double mutant.¹⁵³ Remodelling of the ER occurs during HCV infection,^{154,155} while double-stranded RNA replicative intermediates associate with budding LDs, likely together with core.^{153,156} Genotype-specific substitution of alanine by valine at residue 147 (A147V) within D2 has been observed to decrease mobility of the core protein on LDs and to promote viral assembly and release.¹⁵⁰ Importantly, expression of core protein alone is sufficient to induce budding from ER membranes adjacent to LDs.¹⁵⁷ A genotype 3-specific substitution of tyrosine by phenylalanine at residue 164 (Y164F) within D2 increases LD area in cells producing genotype 1a core protein modified with Y164F inserted into a Semliki Forest virus replicon vector.¹⁵⁸ Finally, Clément et al.¹⁵⁹ have reported that seipin overexpression, which increases LD size and decreases LD number but has no effect on cellular CE or triglyceride levels in Huh-7 hepatoma cells (with the inferred net effect of reducing cellular LD surface area), decreases production of infectious virus, but not replication of viral RNA, for the intragenotypic chimeric virus Jc1,¹⁵⁰ suggesting that available surface area of the hepatocyte LD determines the efficiency of viral assembly. Based on their findings and citing the previous results of Szymanski et al. in human fibroblasts from a patient deficient in BSCL2/seipin,¹⁶⁰ Clément et al.¹⁵⁹ also affirm an important, possibly cell-type-specific, role

for seipin in LD fusion, particularly under normal conditions without lipid overload.

Studying HCV proteins may provide important insights into determinants of LD localisation and routes of access to the LD for LD-associated proteins. NS5A is a viral-encoded RNA-binding phosphoprotein important for both HCV replication and assembly that can localise to and function on host LDs.^{161,162} Host diacylglycerol acyltransferase 1 (DGAT1) promotes the localisation of core and NS5A in complex to the LDs, dependent on the integrity of DGAT1 catalytic activity, via a transient interaction with core and NS5A.^{161,163} Unlike DGAT2, DGAT1 is not localised to LDs, and depletion of either DGAT1 or DGAT2 alone in Huh7.5 hepatoma cells does not affect LD numbers or size, while co-inhibition reduces LD numbers and area.^{161,163} However, assembly of infectious HCV virions is dependent on DGAT1.¹⁶³ Additionally, Vogt et al.¹⁶⁴ and Ploen et al.¹⁶⁵ have reported an interaction between NS5A and the PAT family protein TIP47/PLIN3. Ploen et al. demonstrated that the PAT domain of TIP47 is involved in the interaction with NS5A, while Vogt et al. determined that a mutation, W9A, on the hydrophobic face of the NS5A N-terminal amphipathic alpha helix reduced interaction between TIP47 and NS5A and decreased abundance of both proteins in LD-rich lysate fractions in HCV-infected cells, whereas TIP47 is normally induced to traffic to LD membranes upon HCV infection. While replication of HCV RNA primarily occurs in the membranous web of remodelled ER, HCV RNA and replication complex components NS5A, NS5B and NS3 also occur in LD fractions,¹⁶⁴ and expression of TIP47 is important for HCV particle production and RNA replication,^{164,165} suggesting that LDs are closely associated with the membranous web even in subgenomic replicons missing core with functional roles in the HCV life cycle.¹⁶⁴ Mutations in domain 1 of NS5A, which is C terminal to the alpha helix, alter LD morphology in HCV-infected cells and affect colocalisation with core.¹⁶² Interestingly, an LD-associated host protein viperin, encoded by a type I interferon-stimulated gene, shows antiviral activity against HCV dependent on both its C terminus, which mediates interaction with NS5A and core, and an N-terminal alpha helix similar to that of NS5A that promotes LD localisation in both proteins.^{166–167} A summary of the function of LDs in the viral life cycle is shown in Figure 4A and known determinants of LD association for core and NS5A are shown in Figure 4B.

Proteins located on the LD can have multiple functions and may even be toxic if dysregulated or redistributed away from the LD. Overexpression of the LD-associated host protein cell death-inducing DFF45-like effector B (CIDEb) can reduce proliferation of HCV in Huh7.5 hepatoma cells.¹⁶⁸ CIDEb is a polyfunctional protein that not only stimulates apoptosis when localised to mitochondria in a dimerised form¹⁶⁹ but also promotes LD fusion under conditions of exogenous fatty acid overload,¹⁷⁰ with both processes dependent on nearly overlapping or overlapping sequences in the C terminus of the protein.^{168,171} Gong et al.¹⁷¹ identified a motif in an amphipathic alpha helix containing three basic residues—K/RyxxK/RyxxR, with x representing hydrophobic residues and y representing

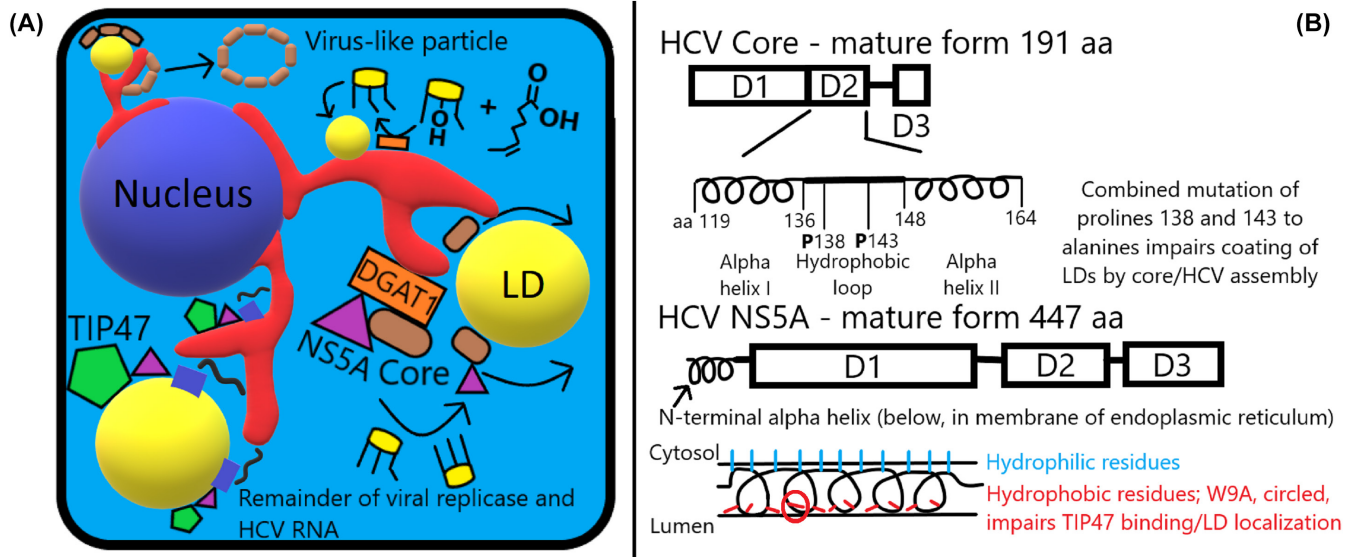


FIGURE 4 (A) Hepatitis C virus (HCV) is strongly dependent on host lipid droplets (LDs) to mount a productive infection. The ER has been modified into a membranous web (red) closely associated with LDs to support infection.^{154,155} Expression of core protein alone is sufficient to induce budding of viral-like particles from sites on the ER adjacent to LD contacts.¹⁵⁷ Host diacylglycerol acyltransferase 1 (DGAT1) helps non-structural protein 5A (NS5A) and core interact and load onto LDs,¹⁶¹ reportedly dependent on the integrity of its enzymatic activity.^{161,163} The kinetics of core movement on LD surfaces may explain differences in viral production between viral strains/genotypes, with a greater degree of photobleaching and slower recovery during fluorescence recovery after photobleaching (FRAP) observed for the core A147V mutation and associated with increases in viral production.¹⁵⁰ Host tail-interacting protein of 47 kDa (TIP47), also known as perilipin 3 (PLIN3), distributes onto LDs following infection by viral replicons and associates with NS5A, which forms a replication complex together with RNA polymerase non-structural protein 5B (NS5B) and NS2, NS3/4A and NS4B to replicate HCV RNA.¹⁶⁴ NS5A may be able to associate with LDs and TIP47 in the absence of core.¹⁶⁴ (B) Two viral proteins of HCV demonstrate distinct features that promote association with LDs. Core protein contains two amphipathic alpha helices in its second domain that are separated by a hydrophobic loop. Two proline residues within the hydrophobic loop are analogous to the 'proline knot' of the plant LD-associated protein oleosin¹⁴⁹ and are together necessary for core to coat the LD. If the residues are mutated to alanines, core remains localised in a punctate distribution, possibly to the ER-LD junction.¹⁵¹ By contrast, NS5A possesses an N-terminal alpha helix that is necessary for its ER/LD distribution.¹⁶⁴ The alpha helix is expected to intercalate into the ER membrane horizontally, with hydrophilic residues (blue) oriented towards the cytosol and hydrophobic residues (red) embedded in the membrane.¹⁶⁴ The mutation of tryptophan to alanine at position 9 within the NS5A alpha helix results in impairment of TIP47 binding and decreases the LD association of both proteins.¹⁶⁴ An N-terminal alpha helix is also required for the LD localisation of host viperin antiviral protein (not shown).¹⁶⁶ Domains II and III of NS5A are intrinsically disordered, while domain I, C terminal to the alpha helix, is highly structured and binds zinc.¹⁶²

hydrophilic residues—conserved in the three members of the CIDE family and required for CIDE/Csp27-mediated growth of LDs by directional lipid transfer from smaller to larger LDs but not the protein's localisation to LD–LD junctions. Interestingly, overexpression of *CIDEB*, in addition to decreasing replication of HCV and causing reduction in cellular triglyceride content associated with fusion of existing LDs, decreases hepatocyte viability.¹⁶⁸ Singaravelu et al. demonstrated that the effects of *CIDEB* on HCV replication were independent of both cellular caspase 3 activity and mutation of the three-basic-residue motif to impair LD fusion.¹⁶⁸ They speculate that a cytopathic effect independent of caspase 3 activity is mediated by an N-terminal DNA fragmentation factor (DFF)-like sequence. CIDE family members, among which *CIDEB* is most abundantly expressed in liver tissue, contain both a CIDE-N (closer to N terminus) domain homologous to those of DFFs and a CIDE-C (closer to C terminus) domain, (not to be confused with the CIDE/Csp27 protein) that is unique to *CIDEA*, *CIDEB* and *CIDEC*.¹⁷² Surprisingly, the CIDE-N domain mediates helical oligomerisation

of DFF proteins by a unique electrostatic mechanism¹⁷³ and at least mediates homodimerisation in *CIDE/Csp27*, important for that protein's ability to transfer lipids between LDs,¹⁷⁴ while the LD localisation of *CIDE/Csp27* is likely promoted by an element within the CIDE-C domain.¹⁷⁵

3.7 | Intracellular lipid droplets

Hepatocytes can also contain intracellular LDs (Figure 2A), some of which appear to be cytoplasmic in origin due to continuity with the cytoplasm on serial electron microscopy sections, either existing within an invaginated nuclear envelope (associated with type 2 nucleoplasmic reticulum or type 2 NR) or within the nucleus proper,¹⁷⁶ presumably due to LD biogenesis within the nuclear envelope (type 1 NR). Downregulation of seipin causes focal accumulations of phosphatidic acid in the nuclear envelope and increases nuclear LDs.¹⁷⁷ Lipin-1, a protein with phosphatidic acid phosphatase activity involved in TAG

and phospholipid biosynthesis,⁹⁶ is important for LD biogenesis from the nuclear envelope in U2OS osteosarcoma cells, while seipin is apparently non-essential.¹⁷⁷ Nuclear LD formation may instead involve a protein involved in the biogenesis of promyelocytic leukaemia (PML) nuclear bodies, PML-II,¹⁷⁸ to which nuclear LDs containing lipin-1, CTP:phosphocholine cytidyltransferase alpha (CCT α), an enzyme that helps generate phosphatidylcholine required for LD biogenesis while localised to the nuclear envelope and cytoplasm but not cytoplasmic LDs,¹⁷⁹ and DAG associate.^{177,180} Intriguingly, lipin-1 nuclear localisation in NIH 3T3 preadipocytes and AML12 hepatocytes is driven by mTOR inhibition, and nuclear lipin-1 appears to assist in inhibition of SREBP when mTORC1 is blocked.¹⁸¹ However, mechanisms of nuclear LD formation¹⁷⁸ and regulation of lipin-1 localisation by mTORC appear to be cell-type specific,¹⁸¹ complicating interpretation of results. Nevertheless, nuclear LDs are a common finding by electron microscopy in CLD according to Imai et al.,¹⁸² with the presence of LDs in association with type 1 NR correlating with markers of ER luminal size and ER stress, while type 2 NR-derived LDs were found in patients with lower serum total cholesterol and LDL levels.¹⁸²

4 | CLDs ASSOCIATED WITH HEPATIC STEATOSIS (SLDs)

4.1 | Hepatic steatosis induced by excessive alcohol ingestion

Ethanol ingestion can represent an acute and/or a chronic stressor to liver metabolism, depending on individual patterns of alcohol consumption. Hepatic steatosis is present in virtually all people who consume more than 40 g of alcohol per day on a regular basis.¹⁸³ Ethanol is metabolised by the sequential action of alcohol dehydrogenase (ADH) and acetaldehyde dehydrogenase (ALDH), leading to a decrease in the NAD⁺/NADH ratio.¹⁸⁴ Chronic alcohol consumption induces microsomal metabolic pathways such as that of CYP2E1, which generate reactive oxygen species.^{73,184} The decreased ratio of NAD⁺/NADH results in product inhibition of 3-hydroxyacyl-CoA dehydrogenase, an enzyme of beta oxidation.¹²¹ Additionally, perturbations in NAD⁺ availability in ALD may affect the function of sirtuins,¹⁸⁴ a family of NAD⁺-dependent deacylases.¹⁸⁵ Sirtuin 1 normally inhibits the lipogenic transcription factors SREBP-1c and carbohydrate-responsive element-binding protein (ChREBP)¹⁸⁶ and supports mitochondrial function through activation of PGC-1 α .¹⁸⁷ Broadly diminished expression of sirtuin 1 has been reported in association with the metabolic syndrome, in patients with psoriasis,¹⁸⁸ and with hepatic steatosis of high grade, in patients with severe obesity.¹⁸⁹ A genetic polymorphism in *ALDH2*, rs671, has been reported to be associated with the presence of hepatic steatosis and NAFLD,¹⁹⁰ although it may protect against alcohol drinking due to loss of function in the encoded enzyme and consequent unpleasant alcohol metabolite-mediated side effects such as flushing and headaches.¹⁹¹ Lipophagy is upregulated in alcohol-induced liver injury and is dependent on SQSTM1,¹⁹² also known as p62, which

accumulates in the MDBs of hepatocytes undergoing ballooning degeneration in ASH and MASH.¹⁹³ Increased de novo lipogenesis, increased peripheral lipolysis, increased esterification of fatty acids into triglycerides and decreased triglyceride secretion are all linked to the development of hepatic steatosis in ALD.^{121,194} The metabolism of ethanol contributes to LD accumulation in hepatocytes, related to decreases in fat efflux that are abrogated by treatment with the ADH inhibitor 4-methylpyrazole.¹⁹⁵

Differences in alcohol metabolism may underlie sex differences in susceptibility to ALD. Although high levels of alcohol consumption are more common in men than women in the United States,¹⁹⁶ women who drink alcohol heavily are at risk of having ALD which progresses to cirrhosis more rapidly than in men.¹⁹⁷ The mechanism underlying this disparity is not clear, as oestrogen appears to be protective against fibrosis development, but is potentially attributable to sex differences in gastric ADH activity, which is higher in men than women and may lead more alcohol to be metabolised by the liver in women.^{198,199} Additionally, alcohol distributes into total body water after absorption, and due to their higher relative proportion of adipose tissue and smaller size, women have, on average, a smaller volume of distribution of alcohol than men do and thus develop relatively greater alcohol concentrations at a similar level of consumption.²⁰⁰ Total body water also decreases with ageing, beginning in the 7th decade of life.²⁰¹

4.2 | Hepatic steatosis in MASLD

Long-term clinical studies have affirmed the progressive nature of MASLD,²⁰² including one paired biopsy study with a median of 11.7 years of subsequent follow-up that showed a 65% increased relative risk of developing decompensated liver cirrhosis, HCC or receiving a liver transplant among people with baseline non-cirrhotic NAFLD and any evidence of incident steatohepatitis, fibrosis or fibrosis progression at a second biopsy more than 6 months after the first.²⁰³ Additionally, people with steatohepatitis are at risk of developing hypertension and type 2 diabetes.²⁰⁴ Careful management of cardiovascular risk factors is needed to prevent coronary heart disease and stroke among patients with MASLD. In this regard, a population-based study by Moon et al.²⁰⁵ found that the odds of cardiovascular disease events were elevated in MASLD, with further increases observed in MetALD and ALD. A large community-based cohort followed by Xu et al.²⁰⁶ demonstrated an elevated risk of incident ischaemic stroke and myocardial infarction in patients with NAFLD independent of metabolic comorbidity that increased with the severity of ultrasound-determined hepatic steatosis, although the presence of liver inflammation or fibrosis was not assessed. Hepatic steatosis may regress during the progression of MASLD to liver fibrosis, an event associated with decreased hepatic fatty acid transport protein 5 (*FATP5*) expression in the setting of lobular inflammation and hepatocellular ballooning per Enooko et al.²⁰⁷ It is not clear whether baseline severity of hepatic steatosis alone affects risk of progressive fibrosis in MASLD.²⁰²

4.3 | Hepatic steatosis in hepatitis C infection

As previously discussed, HCV is intricately dependent on cellular LDs for successful infection, and hepatic steatosis is a common sequela of genotype 3 infection. HCV itself manipulates hepatic lipid metabolism to promote steatosis. It has recently been suggested by Abomughaid et al.²⁰⁸ that accumulation of phosphatidylcholine (PC), a component enriched in LD limiting membranes²⁰⁹ (p. 246), in HCV-infected hepatocytes may be due to viral induction of the phosphatidylethanolamine N-methyltransferase (PEMT) alternative PC synthesis pathway. In addition, a trend towards downregulation of microsomal triglyceride transfer protein (MTP), a chaperone that facilitates lipoprotein secretion, was observed by the authors in HCV-infected cells, with a significant increase following PEMT knockdown,²⁰⁸ suggestive, as are clinical data demonstrating decreased serum cholesterol in CHC patients,⁵⁷ that the virus interferes with lipoprotein secretion from the liver in chronic infection.²¹⁰ Moreover, the single nucleotide polymorphism (SNP) *MTTP*-493G/T (rs1800591) may be associated with more severe steatosis in CHC,²¹⁰ and heterozygosity for the minor allele (T) has synergistic effects on susceptibility to hepatic steatosis in HCV genotype 3 infection.²¹¹ The minor allele increases transcription of *MTTP*, which encodes MTP.²¹¹ To resolve this apparent discrepancy with the known inhibitory effect of HCV core protein on MTP,²¹² Mirandola et al.²¹⁰ propose that upregulation of *MTTP* benefits viral replication early in infection, while chronic inhibition of MTP contributes to viral persistence. Other *MTTP* variants that have been reported to contribute to hepatic steatosis in genotype 3 infection include -164T/C (rs1800804) and -400A/T (rs1800803).²¹³ Hepatic SREBP is also upregulated in chronic HCV infection,²⁰⁸ corresponding to increases in de novo lipogenesis that are essential for HCV replication and release, according to Narayanan et al.²¹⁴

Many cases of hepatic steatosis in CHC, however, especially in non-genotype 3 HCV infections, may be attributable to metabolic factors shared with those that define MASLD. It is believed that HCV can contribute to insulin resistance as a direct viral effect⁵⁴ (p. 424). This is particularly evident in non-genotype 3 infections, such as in people with CHC due to genotype 1 or 4, in whom the presence of insulin resistance has been associated with a high viral load and high-grade (>30%) hepatic steatosis.²¹⁵ Infection of immortalised human hepatocytes with JFH1 strain HCV, the genotype 2a derivative previously discussed, results in downregulation of the glucose transporter GLUT4 and upregulation of gluconeogenic enzyme phosphoenolpyruvate carboxykinase 2 (PCK2).²¹⁶ Multiple mechanisms for HCV-induced insulin resistance, including direct mediation by core protein, have been proposed.²¹⁷

The cooccurrence of hepatic steatosis with CHC is clinically significant. When present, hepatic steatosis may influence the rate of fibrosis progression in CHC,²¹⁸ although Zeremski et al.²¹⁹ have reported results of a serial biopsy study with a mean follow-up period of 6.5 years demonstrating that an initial finding of hepatic steatosis predicts increased stage of fibrosis and degree of inflammation at

baseline but not their progression. The presence of hepatic steatosis does not appear to affect high rates of achievement of virologic cure (sustained virological response at 12 weeks after end of treatment or SVR12) with the new direct-acting antiviral agents (DAAs).²²⁰ The burden of SLD among HCV patients increased during treatment with DAAs, despite achievement of virologic cure, in a recent real-world cohort study by Trifan et al.,²²¹ and those with SLD were observed to have a higher burden of advanced fibrosis. Additional prospective data on outcomes of persistent and new-onset SLD after SVR12 are needed to determine need for special screening and follow-up in this population.²²²

4.4 | Hepatic steatosis in genetic diseases

Genetic conditions associated with hepatic steatosis and progressive liver disease include Wilson's disease, other inborn disorders of metabolism, familial hypobetalipoproteinaemia⁹⁹ and certain congenital disorders of glycosylation.²²³ Wilson's disease, caused by mutations in the *ATP7B* gene and associated copper accumulation, is frequently accompanied by macrovesicular steatosis that is panlobular in distribution and, as is often seen in MASLD, may be accompanied by the histologic finding of glycogenated nuclei.²²⁴ Hepatic steatosis in Wilson's disease is additionally closely associated with mitochondrial ultrastructural abnormalities, which occur early in the disease process.²²⁵ Due to variable penetrance, Wilson's disease can manifest at any age and liver involvement is often the earliest finding.²²³ Acute liver failure is an uncommon initial presentation of Wilson's disease.²²³

Familial hypobetalipoproteinaemia is also an important monogenic contributor to SLD. People who are heterozygous for loss-of-function mutations in *APOB* frequently have liver steatosis with a significant proportion, reportedly 5%–10%, developing steatohepatitis.²²⁶ Patients with biallelic pathogenic mutations (compound heterozygous or homozygous) are at risk of more severe liver-related manifestations, accompanied by steatorrhea, failure to thrive and neurologic complications necessitating a low-fat diet and fat-soluble vitamin supplementation.²²⁶ Abetalipoproteinaemia, an autosomal recessive disorder caused by loss-of-function mutations in *MTTP*, presents early in life and a subset of patients may develop advanced fibrosis despite low BMI.²²⁷

Defects in adipocyte fat storage, including partial and total lipodystrophies, can also be a rare cause of hepatic steatosis. A homozygous mutation in *CIDEA*, encoding the previously mentioned LD-associated CIDEA/Fsp27 protein, has been described in a case report²²⁸ as a cause of partial lipodystrophy, diabetes, dyslipidaemia and hepatic steatosis. The mutation, p.E186X, results in a premature stop codon in the CIDE-C domain that prevented CIDEA LD localisation when it was expressed in adipocyte cell cultures and reduced LD expansion during lipid load in pre-adipocytes and fibroblast-like cells.²²⁸ CIDEA/Fsp27 is not significantly present in liver, where CIDEB is predominant,¹⁷² and thus, the ability to develop fatty transformation of hepatocytes is spared, although it is not clear if they

experience alterations in LD morphology. Rubio-Cabezas et al.²²⁸ note that lipodystrophy is a distinct phenotype that results in a positive energy balance due to inability to form adipose tissue fat stores, frequently causing insulin-resistant diabetes in these patients. Additionally, adipokines, a type of endocrine signal released by adipocytes, may be deficient in lipodystrophies, while dysregulation of important adipokine signalling pathways, such as those of leptin and adiponectin, has been linked to disease progression in MASLD, CHC and chronic hepatitis B.²²⁹

In cases of idiopathic and severe hepatic steatosis, genetic testing has clinical utility. Hakim et al.²³⁰ describe whole-exome sequencing in a group of 19 individuals with unexplained CLD, 6 of whom were characterised as having non-obese NAFLD with or without steatohepatitis. Monogenic disorders were identified in five patients, including, among the non-obese NAFLD subjects, an individual with familial partial lipodystrophy type 3, a person with mitochondrial complex I deficiency and another with heterozygous familial hypobetalipoproteinaemia. These patients were able to receive targeted therapies for their respective conditions. In addition to monogenic disorders, it should also be recognised that polygenic disorders resulting from epistasis and pleiotropy can contribute to cases of chylomicronaemia syndrome, hypertriglyceridaemia and possibly metabolic syndrome more broadly,²³¹ and the role of epigenetics, including histone modification and microRNA regulation, in these disorders is also important and understudied.²³²

4.5 | Hepatic steatosis in drug-induced liver injury

Drug-induced liver injury can also cause chronic SLD and, in certain cases, steatohepatitis (Table 1). Kleiner⁴ (pp. 747–748) lists 134 chemicals, drugs, toxins and herbal medications known to be associated with steatosis and/or steatohepatitis. Methotrexate, an immunosuppressant that antagonises dihydrofolate reductase, is a known cause of hepatic steatosis and steatohepatitis that mimics MASLD. The mechanism by which methotrexate induces steatosis is not known but is likely polyfactorial, including the depletion of S-adenosylmethionine and associated effects on one-carbon metabolism-dependent very-low-density lipoprotein (VLDL) secretion, as well as impairment of mitochondrial respiratory chain function.²³³ Tamoxifen, a selective oestrogen receptor modulator, can induce steatosis both through increased de novo lipogenesis associated with pregnane X receptor activation and by impairment of mitochondrial function.²³³ As a cationic drug, tamoxifen enters mitochondria by an electrophoretic mechanism and can induce mitochondrial permeability transition pores within the inner mitochondrial membrane, a calcium-dependent process.²³³ Long-term therapy can result in chemotherapy-induced steatohepatitis (CASH),²³⁴ which was observed by Saphner et al.²³⁵ to be reversible following tamoxifen discontinuation in 14 of 16 patients.

Chemotherapy-induced steatohepatitis can be aetiologically related to specific chemotherapy agents.²³⁶ In addition to methotrexate and tamoxifen, 5-fluorouracil, irinotecan and L-asparaginase

therapy are risk factors for hepatic steatosis and/or CASH.²³⁴ Irinotecan is an important cause of iatrogenic morbidity and mortality related to CASH in people being treated for colorectal cancer liver metastasis.²³⁶ Irinotecan interferes with autophagy by altering lysosomal pH and by indirect activation of p38 mitogen-activated protein, which inhibits autophagosome development.²³⁶ Irinotecan additionally inhibits fatty acid beta oxidation and stimulates release of reactive oxygen species from mitochondria in hepatocytes.^{236,237} A summary of causes of SLD pathogenesis is shown in Figure 5.

4.6 | Hepatic steatosis in patients with HIV infection

People living with HIV/AIDS are a particularly important and understudied population with respect to outcomes of SLD. Weight gain is an expected consequence of initiation of antiretroviral therapy as part of a return to health. However, weight gain among patients living with HIV was observed in a large study by Bischoff et al.,²³⁸ particularly in men, those with type 2 diabetes or with a nadir CD4 count less than 200 per microliter, to be associated with new development of hepatic steatosis. Steatosis was present at the end of a mean 41 months of follow-up in 50% of this largely non-obese population.²³⁸ Specific antiretroviral agents used in combination therapy, such as integrase strand transferase inhibitors, may be associated with greater weight gain and hepatic steatosis,²³⁸ perhaps related to effects on central nervous system melanocortin signalling.²³⁹ A recent study in the United States by Gawrieh et al.²⁴⁰ among people with chronic HIV identified clinically significant hepatic fibrosis by transient elastography in 21% of persons with MASLD and 15% of the total study population. Treatment of MASLD in people with HIV/AIDS is complicated by lack of study in this population but given the potential benefit of lifestyle modifications and rational antiretroviral medication substitutions, screening for MASLD among patients with HIV/AIDS should be strongly considered.²⁴¹

5 | HEPATIC STEATOSIS IN ACUTE LIVER INJURY AND FAILURE

Acute liver failure is a feared condition that creates diagnostic challenges. Hy's law estimates a mortality rate of 10% in patients with drug-induced liver injury and evidence of hepatocellular injury and jaundice.²⁴² Alternatively, a predictive score for acute liver failure in DILI, developed by Lo Re et al.,²⁴³ uses platelet count and bilirubin to optimise sensitivity. Acute liver failure accompanied by hepatic steatosis is associated with multiple hepatotoxic medications and toxic substances, acute fatty liver of pregnancy, Reye's syndrome as well as congenital causes (Table 2). Microvesicular steatosis, thought to be due to severe impairment of mitochondrial beta oxidation, may be seen in acute liver injury.

Intravenous high-dose amiodarone and tetracycline are two potential causes of acute liver failure associated with development of

TABLE 1 Causes of chronic liver disease (CLD) associated with hepatic steatosis (steatotic liver disease, SLD).

Aetiology	Pathologic associations
<i>Dietary or metabolic</i>	
Choline deficiency	- Risk factors include male sex and post-menopausal status - De novo synthesis of choline by phosphatidylethanolamine <i>N</i>-methyltransferase is controlled by gene expression of <i>PEMT</i>, influenced by an oestrogen response element³⁴¹
Chylomicronaemia syndrome	- Familial hyperchylomicronaemia is associated with mutations in <i>LPL</i> and more rarely <i>APOC2</i>, <i>APOA5</i>, <i>GPIHBP1</i> and <i>LMF1</i>, while multifactorial chylomicronaemia syndrome is polygenic, involving these genes and others, such as <i>CREB3L3</i> - Chylomicronaemia syndrome is associated with diabetes, hypertriglyceridaemia and hepatic steatosis²³¹
Hereditary fructose intolerance ³⁴³	SLD may be seen in patients in treatment independent of obesity or insulin resistance³⁴²
Severe malnutrition	- Association of fatty liver with Kwashiorkor disease³⁴³ - Adults who have past medical history of childhood severe wasting may be at risk of SLD associated with obesity⁵¹
Metabolic dysfunction-associated steatotic liver disease (MASLD, formerly known as non-alcoholic fatty liver disease)	Predominantly macrovesicular steatosis, although microvesicular changes can be seen in approximately 10% of NAFLD biopsies³⁴⁴
<i>Infectious</i>	
Hepatitis C viral infection	- A member of the Flaviviridae family - Strongest association of hepatic steatosis with genotype 3-associated chronic hepatitis C^{57,145,146}
<i>Monogenic</i>	
Abetalipoproteinaemia	- Defect in <i>MTTP</i> gene; associated with acanthocytes - Treatable with low-fat diet and vitamin A and E supplementation; medium-chain triglyceride supplementation is no longer recommended due to its association with liver disease progression; and liver transplant can normalise serum lipids but not fat malabsorption⁴⁴ (pp. 194–195)
Bardet-Biedl syndrome, Alström syndrome	Ciliopathies, causes of paediatric steatohepatitis associated with insulin resistance and increased circulating insulin²⁶ (p. 332);²⁶ (p. 332)
Bloom syndrome	- Mutations in <i>BLM</i> gene - Case report of sclerosing hyaline necrosis of the liver, otherwise characteristic of alcohol-related liver disease or ALD²⁶ (p. 345), in an adult patient with Bloom syndrome³⁴⁵
Chanarin–Dorfman syndrome (neutral lipid storage disease)	- Associated with mutations in <i>ABHD5</i> and <i>PNPLA2</i> genes - Variable severity of liver involvement,³⁴⁶ and micronodular cirrhosis may develop in childhood³⁴⁷ or later - Jordans' anomaly, lipid droplets present in granulocytes in peripheral blood smear³⁴⁶
Congenital total lipodystrophy	- Severe syndrome with lack of adipose tissue, hyperlipidaemia, muscle hypertrophy, hypertrichosis, acanthosis nigricans and diabetes mellitus - Associated with mutations in <i>AGPAT2</i>, <i>BSCL2</i>, <i>CAV1</i> and <i>PTRF</i> genes⁴⁴ (p. 230)
Cystic fibrosis	Often periportal steatosis; steatosis degree does not appear to correlate with nutritional status; and focal biliary fibrosis common⁴⁴ (p. 155)
Familial hypobetaproteinaemia	Associated with mutations in <i>APOB</i> gene⁴⁴ (pp. 194–195)
Familial partial lipodystrophy	- Most common subtype of this rare disease is associated with mutations in <i>LMNA</i> gene³⁴⁸ - Hypertriglyceridaemia and hepatic steatosis are common²³¹
Lysosomal acid lipase deficiency (cholesteryl ester storage disease and Wolman disease)	- Enlarged Kupffer cells and hepatocyte appearance that resembles small-droplet macrovesicular or microvesicular steatosis, however, vacuoles are positive for cathepsin D lysosomal marker^{41,42} and LAMP,⁴² therefore they are not true lipid droplets - Birefringent cholesterol crystals may be observable in fresh tissue⁴¹ - Accompanied by hyperlipidaemia and elevated chitotriosidase^{104,349} - Autosomal recessive mutations in <i>LIPA</i> gene; Wolman disease is the most severe form that presents in infancy with acute liver injury⁴⁴ (pp. 195–197), associated with adrenal gland calcifications¹⁰⁴
Werner syndrome	- Associated with mutations in <i>WRN</i> gene causing premature ageing - Insulin resistance and central obesity may be associated with steatohepatitis in these patients³⁵⁰

TABLE 1 (Continued)

Aetiology	Pathologic associations
Wilson's disease (hepatolenticular degeneration)	- Hepatic copper accumulation, neuropsychiatric symptoms, Kayser–Fleischer rings; variable penetrance; and liver disease is often the earliest manifestation, associated with decreased activity of liver X receptor–retinoid X receptor complex - Uncommonly causes acute liver failure²²³
<i>Toxin or drug mediated</i>	
ALD	- Steatosis is the earliest lesion of ALD, and may be associated with foamy degeneration, Mallory–Denk bodies and steatohepatitis - Presence of giant mitochondria and mixed macrovesicular and microvesicular fat reported to predict progression to fibrosis and cirrhosis³⁵¹ - Treatment involves supporting alcohol abstinence with nutritional support²⁶ (p. 352)
Amiodarone	- May have both macrovesicular and microvesicular steatosis²⁴⁶; Mallory–Denk bodies and pericellular fibrosis in zone 1⁵⁶ - Intravenous administration is rarely associated with acute liver injury⁴ (pp. 749, 800);^{245,246}
Cadmium ³⁴³	Urinary cadmium excretion, associated with tobacco smoking, an important source of cadmium exposure, and age, has been independently associated with hepatic steatosis in men³⁵²
4,4'-Diethylaminoethoxyhexestrol	Findings include hyperlipidaemia, phospholipidosis and elevated hepatic cholesterol,³⁵³ in some cases with increased hepatic triglycerides³⁵⁴ and/or Mallory–Denk bodies²⁴⁶
Diethylstilboestrol	- Synthetic oestrogen used in prostate cancer therapy - Long-term high-dose therapy associated with pericentral steatosis with ballooned hepatocytes, resembling alcohol-related hepatitis⁴ (p. 749);³⁵⁵ and peliosis hepatis⁴ (p. 755)
Irinotecan	- Derivative of plant-derived camptothecin - A cause of chemotherapeutic agent-associated steatohepatitis²⁶ (p. 378);²³⁶
Lead ³⁴³	Hepatic triglyceride content was found to be associated with cumulative childhood blood lead levels in young adult patients who had annual blood lead level monitoring from age 1 to 4 years³⁵⁶
Methotrexate ⁴ (p. 793)	- Promotes steatosis through polyfactorial mechanisms²³³ - Risk factors for liver injury include obesity, long-term alcohol use and hepatitis C³⁵⁷ - Can cause acute injury when used in large doses for cancer therapy⁴ (p. 804)
Perhexiline maleate	- Associated with macrovesicular steatosis⁴ (p. 747) and phospholipidosis⁴ (p. 747);³⁵⁸ - Early mechanisms of injury include ER stress and p38 activation³⁵⁹; may inhibit carnitine palmitoyltransferase-1 - Toxicity results from long-term therapy of 3 weeks to 3 years duration that may be avoided by identifying poor metabolisers (with CYP2D6 polymorphism)^{360,361} and monitoring drug serum levels³⁶²
Tamoxifen ⁴ (p. 798); ²⁶ (pp. 357–358)	A2 allele of CYP17A1 is a risk factor for hepatic steatosis in women with tamoxifen treatment for breast cancer^{234,363}
Tetrachloroethylene (perchloroethylene)	Chronic exposure to the solvent may be associated with central nervous system, dermatologic and liver effects, possibly including toxicant-associated steatohepatitis³⁶⁴ and SLD³⁴³
Valproic acid	- Long-term therapy is associated with chronic SLD and with increased expression of PPARγ and CD36, but without effect on carnitine palmitoyltransferase Ia gene expression²⁵⁵ - Can also be a cause of acute hepatotoxicity²⁵⁴
Vinyl chloride monomer	Retrospective study in workers with very high levels of exposure (~1000 ppm/day) prior to implementation of occupational exposure limits found 84% steatosis prevalence and 80% steatohepatitis prevalence in 25 non-obese workers, 4 of whom had hemangiosarcoma and 1 of whom developed hemangiosarcoma during follow up; 76% had hepatic sinusoidal dilation; steatohepatitis was not associated with transaminitis in this population³⁶⁵
Volatile petrochemicals	Non-alcoholic steatohepatitis identified in 20 patients working in the petrochemical industry in north-eastern Brazil improved in all 10 removed from their working environment³⁶⁶

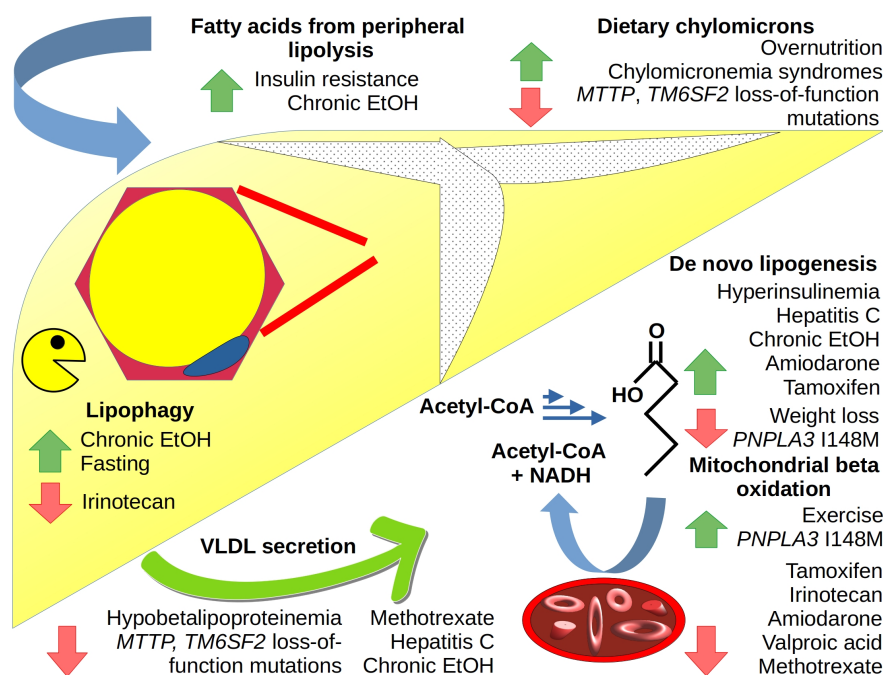


FIGURE 5 Lipid anabolic and catabolic pathways within the liver associated with development of chronic steatotic liver disease. EtOH, ethanol; MASLD, metabolic dysfunction-associated steatotic liver disease; MTTF, microsomal triglyceride transfer protein; PNPLA3, patatin-like phospholipase domain-containing 3; TM6SF2, transmembrane 6 superfamily member 2.

hepatic steatosis. Amiodarone and tetracycline both upregulate expression of *SREBP1* and *PPARG*, genes with lipogenic protein products, in HepaRG cells²⁴⁴ and are also known to inhibit mitochondrial beta oxidation.⁷³ Tetracycline-induced acute hepatic injury is associated with intravenous administration of more than 1.5 g per day, especially in pregnant women or people with impaired kidney function, associated with microvesicular steatosis⁴ (p. 778). Amiodarone has been described in case reports as a rare cause of acute liver injury when administered intravenously, with polysorbate 80-based formulations possibly implicated⁴ (p. 800).²⁴⁵ Subacute or chronic injury related to amiodarone may exhibit pathological findings of macrovesicular and microvesicular steatosis with MDBs and ballooning, and hepatocyte dysplasia has also been reported.²⁴⁶ Amiodarone-treated cells develop lamellar bodies, a finding associated with phospholipidosis, in addition to LDs²⁴⁴; cationic, amphiphilic drugs like amiodarone may form complexes with phospholipids in lysosomes and thereby inhibit phospholipid metabolism.^{233,247} Drug-induced phospholipidosis may be seen in isolation or association with steatohepatitis depending on the agent⁴ (pp. 747–750)—by contrast, nifedipine might rarely cause subacute²⁴⁸ steatohepatitis without phospholipidosis as an idiosyncratic drug reaction⁴ (pp. 729–730, 749). LDs themselves can be a site of drug metabolism, as was recently shown in the case of lasonolide A, a lipophilic macrolide.²⁴⁹

Valproic acid can also cause microvesicular steatosis and acute liver failure, which has also been reported accompanied by subclinical pancreatitis.²⁵⁰ Valproic acid hepatotoxicity is exacerbated by concomitant cytochrome p450 induction by agents such as phenobarbital, especially in children under 2 years old, suggesting that metabolites of valproate, such as 4-ene-VPA, are mechanistically involved.²⁵¹ As a fatty acid, valproic acid is conjugated to coenzyme A and inhibits carnitine palmitoyl-transferase 1, suggesting that impaired oxidation of fatty acids contributes to the toxidrome.^{252,253}

Cases of valproic acid toxicity causing acute liver failure have been reportedly responsive to L-carnitine therapy,²⁵⁴ consistent with this hypothesis. Valproic acid may also be linked to development of chronic hepatic steatosis. Increased nuclear localisation of PPARgamma (PPAR γ) and cell surface expression of CD36 have been reported in mouse hepatocytes treated with valproic acid, which induced lipotoxicity in combination with exogenous oleic acid.²⁵⁵

Reye's syndrome, an acute non-inflammatory encephalopathy with acute liver failure in children and young adults, appears polyfactorial in origin, associated with endotoxaemia, salicylate toxicity, viral infection and short- and medium-chain fatty acid release into the urine.²⁵⁶ Build-up of methylmalonyl CoA and propionyl CoA may inhibit carbamyl phosphate synthase I and N-acetylglutamate synthase, leading to hyperammonaemia, while hypoglycaemia is associated with a low coenzyme A (CoA-SH)-to-acetyl CoA ratio, the inhibition of pyruvate carboxylase by methylmalonyl CoA and inhibition of citrate synthase by methylmalonyl CoA and propionyl CoA.²⁵⁶ Inborn errors of ureagenesis can also give rise to microvesicular steatosis and acute liver failure in infants, as the ammonium ion inhibits liver fatty acid medium- and long-chain beta oxidation.¹²¹ Acute fatty liver during pregnancy is a life-threatening maternal condition that has been associated with long-chain acyl-coenzyme A dehydrogenase (LCHAD) deficiency in the foetus.^{257,258} The defect is in the mitochondrial trifunctional protein, a hetero-octomer of four alpha and four beta subunits that possesses LCHAD enzymatic activity.²⁵⁷

One example of a cause of toxin-induced acute liver failure is Jamaican vomiting sickness, caused by ingestion of cooked or raw unripe Ackee fruit, its seed or husk or the water used in their boiling. In these cases, it is important to establish the quantity of ingestion and the timing.²⁵⁹ Ripeness is indicated by the colour of the husk and the splitting of the husk to reveal the seeds and aril fruit.²⁵⁹ The toxidrome is characterised by hypoglycaemia, likely due to the action

TABLE 2 Causes of acute liver failure associated with hepatic steatosis.

Aetiology	Pathologic associations
Infectious	
Fulminant hepatitis D caused by hepatitis D virus genotype 3 (Santa Marta hepatitis and Lábreas hepatitis)	• Santa Marta hepatitis is associated with microvesicular steatosis, hepatocellular eosinophilic necrosis and hepatitis D virus infection³⁶⁷ • Lábreas hepatitis is associated with 'morula-like'³⁶⁸ steatosis and has been reported in association with hepatitis D²⁶ (p. 333);³⁶⁸ • Santa Marta hepatitis and Lábreas hepatitis were likely caused by outbreaks of hepatitis D genotype 3 virus in South America³⁶⁹ • Hepatitis D virus (HDV) is a defective RNA virus that depends on the presence of hepatitis B surface antigen for its assembly and secretion³⁷⁰ • Hepatitis D viral infection can cause acute or CLD in people who are coinfecting with hepatitis B virus or cause a superinfection in people who have underlying chronic hepatitis B³⁷¹ • Unknown what virologic, host and/or environmental factors may contribute to the acute liver failure associated with hepatic steatosis
Yellow fever	• Caused by virus of Flaviviridae family, associated with perivenular steatosis, both macrovesicular and microvesicular, Councilman–Rocha Lima bodies and rare intranuclear Torres bodies³⁷² (pp. 450–452) • Hepatic steatosis also associated with dengue fever caused by an arbovirus of the same family as yellow fever virus³⁷² (pp. 452–453)
Mitochondrial	
Pearson's syndrome	• Characterised by large deletions, rearrangements or duplications of mitochondrial DNA that may be variable between tissues³⁷⁵ • Microvesicular hepatic steatosis³⁷⁸ • Refractory sideroblastic anaemia³⁷⁸ • Villous atrophy syndrome has been associated with hepatic steatosis and rearrangement of mitochondrial DNA³⁷⁵
Monogenic	
Acute fatty liver of pregnancy	• Associated with third trimester of pregnancy, but has been reported in second trimester and post-partum • Associated with long-chain 3-hydroxyacyl-coenzyme A dehydrogenase deficiency in foetus (HADHA c. 1528G>C is common, as is compound heterozygosity),²⁵⁷ male foetus, multiple gestation and prior attacks are risk factors²⁵⁸
Hepatocerebral mitochondrial DNA depletion syndromes ³⁷³	• Childhood and infantile forms of Navajo neurohepatopathy are characterised by acute liver failure, while a classic form presents with neurologic symptoms that precede development of cirrhosis • Navajo neurohepatopathy is associated with MPV17 mutations (a nuclear gene encoding a mitochondrial inner membrane protein) and mitochondrial DNA depletion⁴⁴ (pp. 212–213);^{374,375} • POLG mutations (a nuclear gene encoding mitochondrial DNA polymerase gamma) associated with Alpers–Huttenlocher syndrome can similarly result in macrovesicular and microvesicular hepatic steatosis^{373–375} • Use of valproic acid or divalproex can cause serious liver dysfunction in children with Alpers–Huttenlocher syndrome or childhood myocerebrohepatopathy spectrum, which are POLG-related disorders³⁷⁶
Inborn errors of ureagenesis	• Deficiency in urea cycle enzymes including ornithine transcarbamylase or carbamoyl phosphate synthase¹²¹ • Argininosuccinic aciduria is associated with argininosuccinic lyase deficiency, macrovesicular steatosis and megamitochondria⁴⁴ (pp. 191–192)
Wilson's disease (hepatolenticular degeneration)	• More commonly causes CLD, but may present with acute liver injury or failure characterised by intravascular haemolysis, coagulopathy, hepatic encephalopathy, renal failure, low alkaline phosphatase and greater elevation in aspartate aminotransferase (AST) than alanine aminotransferase (ALT)^{223,377}
Toxin or drug induced	
Ackee tree (<i>Blighia sapida</i>), Jamaican vomiting sickness	• Microvesicular steatosis associated with consumption of unripened Ackee tree fruit, hypoglycin A toxin and jaundice⁴ (p. 765)
Alcohol-associated hepatitis	• Neutrophilic infiltrate, necrosis and variable steatosis²⁶ (pp. 343–344), often superimposed on advanced chronic liver disease as acute-on-chronic liver failure in the setting of active alcohol use³⁷⁹ • Stringently selected patients based on clinical and laboratory data benefit from corticosteroids²⁶ (p. 352) • Early transplantation for treatment non-responders shows survival benefit and is the subject of ongoing research³⁸⁰

(Continues)

TABLE 2 (Continued)

Aetiology	Pathologic associations
Arsenic poisoning (inorganic arsenic)	- Reported to cause acute-on-chronic liver failure in non-alcoholic steatohepatitis-associated cirrhosis - As(III) is more toxic than As(V) - Associated with generation of reactive oxygen species and lipid peroxidation³⁸¹
Asparaginase	- Steatosis may be macrovesicular - Short latency of toxicity onset with jaundice³⁸²
Boric acid	Associated with microvesicular steatosis⁴ (p. 762)
Camphor	- Can cause microvesicular steatosis⁴ (pp. 747, 764) - Case report of toxicity associated with chronic camphor exposure in an infant with macrovesicular steatosis and neuronal oedema mimicking Reye's syndrome but without ultrastructural findings of dysmorphic mitochondria³⁸³
Carbon tetrachloride	- Haloalkylation inhibits lipoprotein export - Also associated with lipid peroxidation and defects in calcium homeostasis⁴ (p. 761) - Macrovesicular steatosis⁴ (p. 747) - Trichloroethylene can produce similar findings⁴ (pp. 762–763)
Chlordecone (Kepone)	- Insecticide associated with microvesicular steatosis⁴ (p. 762) - Morbidity more related to neurologic toxicity than hepatotoxicity⁴ (p. 765)
Cocaine ²⁶¹	- Associated with microvesicular steatosis⁴ (p. 746) - Accumulation of hepatotoxic metabolites by CYP2E1 and CYP2A pathways may cause an interaction with concomitant alcohol use disorder⁴ (p. 766)
Didanosine ³⁸⁴	Didanosine and stavudine cause mitochondrial toxicity, alone or especially in combination³⁸⁵
Dimethylformamide ³⁸⁶	Associated with microvesicular steatosis⁴ (p. 746)
Ethionine	Antimetabolite associated with both macrovesicular and microvesicular steatosis⁴ (p. 747)
Fialuridine	Nucleoside analogue associated with diffuse microvesicular steatosis⁴ (p. 782)
Halothane	- Anaesthetic agent that may cause hepatic necrosis and steatosis, metabolised to a trifluoroacetyl halide intermediate⁴ (pp. 761–762) - Epoxide hydrolase deficiency may be a familial risk factor⁴ (p. 731), while prior phenobarbital administration is a clinical risk factor - Liver injury may develop days after a repeat halothane exposure⁴ (p. 812), and immunologic-mediated injury is more common than hepatotoxic metabolite injury; halothane may also cause hypoxic injury⁴ (pp. 784–785)
Hydrazine	Associated with microvesicular steatosis⁴ (p. 762)
Indinavir	- Protease inhibitor - Case reports of toxicity, in one case featuring microvesicular steatosis and eosinophilic infiltrate in patient with underlying cirrhosis⁴ (p. 781);³⁸⁷
Indomethacin	Rare association with acute, severe drug-induced liver injury and microvesicular steatosis⁴ (p. 792)
Interleukin-6	Administration associated with zone 3 injury including ballooning and cholestasis⁴ (p. 807)
Jin Bu Huan and Syo-Saiko-To	- Products in Chinese traditional medicine derived, respectively, from <i>Lycopodium serratum</i> and the root of <i>Scutellaria</i> spp.⁴ (p. 768) - Both associated with microvesicular steatosis⁴ (p. 769) - Other names for syo-saiko-to include xiao-chai-hu-tan and dai-saiko-to⁴ (p. 769)
Linezolid	- Antibiotic - Rare association of prolonged therapy with steatosis and lactic acidosis^{388–389}
Phalloidin	- Mushroom toxin causing steatosis and zone 3 necrosis - Symptoms may be delayed by 6–20 h after ingestion - Associated with actin polymerisation inhibition⁴ (p. 765) and peliosis hepatis⁴ (p. 755)
Reye's syndrome (acute non-inflammatory encephalopathy with fatty liver failure)	- Diffuse microvesicular steatosis, megamitochondria, dysmorphic mitochondria, reduced succinyl dehydrogenase and cytochrome oxidase activity; hyperammonaemia; encephalopathy associated with hypoxia and metabolic derangement - Associated with use of aspirin in children during flu-like illness⁴⁴ (pp. 231–232) - Distinguish from aspirin overdose, which has less striking steatosis and generally some necrosis⁴ (p. 790)
Stavudine	Didanosine and stavudine cause mitochondrial toxicity, alone or especially in combination³⁸⁵

TABLE 2 (Continued)

Aetiology	Pathologic associations
Tetracycline	- High-dose intravenous use associated with acute liver failure - Microvesicular steatosis accompanied by scant or no necrosis and scant cholestasis⁴ (p. 778) - Steatosis often perivenular⁴ (p. 749)
Thallium	- An element - Exposure associated with microvesicular steatosis⁴ (p. 762)
Toluenes and xylenes	- Glue components associated with microvesicular steatosis⁴ (p. 762) - Mortality from acute exposure is typically associated with cardiovascular or neurologic effects rather than hepatotoxicity^{390,391}
Valproic acid	- Acute overdose-related hepatotoxicity may be responsive to L-carnitine therapy²⁵⁴ - A second form of acute hepatotoxicity can develop after 1–6 months of therapy, likely due to a hepatotoxic metabolite, with children under 2 years old being most susceptible⁴ (pp. 788–789) - Chronic therapy can lead to weight gain and development of hepatic steatosis²⁵⁵
White phosphorus	- Microvesicular steatosis, which may be periportal initially⁴ (p. 749), and periportal hepatocyte loss - Gastrointestinal and neurologic toxicity - Malodorous breath (garlic-like) - Phosphorescent stools and vomit⁴ (pp. 763–764)
Zidovudine (azidothymidine)	Mechanism of steatosis and lactic acidosis development involves mitochondrial DNA damage⁴ (p. 781)

of hypoglycin A toxin, accompanied by inhibition of long-chain fatty acid beta oxidation by the hypoglycin metabolite methylene cyclopropyl acetic acid.²⁵⁹ The vomiting that often results induces dehydration. A case report by Grunes et al.²⁶⁰ describes the induction of acute liver failure and red blood cell sickling by ackee fruit poisoning in an individual with previously unknown sickle cell trait. It is important to exclude fulminant viral hepatitis by hepatitis A, B or E viruses, globally common causes of acute liver failure,²⁶¹ in these cases and to establish travel, dietary and medication and supplement history. Treatment is supportive,²⁵⁹ and prompt evaluation for liver transplants can be lifesaving.^{260,262} Further research into the mechanisms of action of hypoglycin A and B toxins is needed to develop specific therapies for Jamaican vomiting sickness.

6 | ROLES OF GENETIC VARIATION IN PROMOTING HEPATIC STEATOSIS

Triglyceride synthesis and storage in the liver are regulated by multiple genetic pathways. Thus, genetic variation can predispose to the development of MASLD and potentially other causes of liver failure. The gene patatin-like phospholipase domain-containing 3 (*PNPLA3*) was identified by Romeo et al.²⁶³ in an American genome-wide association study as harbouring two SNPs associated with liver fat content as assessed by proton MR spectroscopy. The SNPs, rs738409 (p.I148M) and rs6006460 (p.S453I), were associated with increased and decreased liver fat content respectively. Additionally, *PNPLA3* I148M was associated with increased liver inflammation, a finding that was independently discovered in European populations.²⁶⁴ Notably, Romeo et al.²⁶³ found that probable null mutations in *PNPLA3* occurred in subjects with the highest levels of hepatic steatosis, suggesting a loss-of-function mechanism as responsible for the association. The *PNPLA3* enzyme possesses both triglyceride

lipase and acyltransferase activity,²⁶⁵ however, the mechanisms by which its variants may cause or protect from hepatic steatosis are not known at present. Either knockin of the I148M variant or knockout of *PNPLA3* increases the production of LDs in epidermoid carcinoma cells in response to exogenous polyunsaturated fatty acids.²⁶⁶ Increases in hepatic triglycerides containing polyunsaturated fatty acids are accompanied by decreases in hepatic PC in people homozygous for *PNPLA3* p.I148M leading Luukkonen et al.²⁶⁶ to suggest a role for *PNPLA3* in transacylation of unsaturated fatty acids, particularly polyunsaturated ones, into PC and in promoting export of polyunsaturated fatty acids in VLDL particles from the liver, functions that are impaired by the I148M mutation. Luukkonen et al.²⁶⁷ more recently reported that homozygous carriers of *PNPLA3* p.I148M have evidence of decreased Krebs cycle activity, related to mitochondrial dysfunction, which leads to enhanced ketogenesis and impaired hepatic de novo lipogenesis, contrary to the increases in de novo liver lipogenesis typically observed in MASLD.⁴⁵ Structural studies suggest that the replacement of isoleucine by methionine at residue 148 may disrupt substrate binding by *PNPLA3*.¹⁸³ Yuan et al.²⁶⁴ note that the location of rs738409 corresponds to a putative exon splicing silencing site in *PNPLA3*.

Variants in *CIDEA*, which encodes an LD-associated protein that, as previously mentioned, promotes LD fusion, with predicted loss of function are associated with lower occurrence of steatosis, steatohepatitis or fibrosis in people receiving bariatric surgery.¹⁷⁰ Additionally, a donor splice site mutation in *HSD17B13* (rs72613567), causing loss of function and truncation of the encoded LD-associated protein 17-beta-hydroxysteroid dehydrogenase 13, is apparently protective against steatohepatitis but may increase uncomplicated steatosis in bariatric surgery patients.²⁶⁸ The same authors²⁶⁸ also found that rs72613567 was associated with decreased *PNPLA3* expression and was hepatoprotective in people with *PNPLA3* p.I148M. A small case-control study by Faulkner et al.²⁶⁹ identified a variant in

the perilipin *PLIN2*, p.S251P (rs35568725), as linked to non-alcoholic steatohepatitis, with overexpression of the variant leading to altered LD morphology in oleic acid-treated hepatoma cells, like what was previously described in macrophage cell lines.²⁷⁰ The same variant was described as associated with reduced insulin secretion but normoglycaemia in response to oral glucose challenge in a cohort without diabetes carrying rs35568725.²⁷¹

Other gene associations with development of hepatic steatosis include *TM6SF2*, where the rs58542926 variant is associated with higher liver triglyceride content and predisposes to fibrosis progression in NAFLD,²⁷² but is also associated with diminished serum triglycerides and low-density lipoprotein cholesterol.²⁷³ Loss of function of *TM6SF2* reduces lipoprotein secretion and increases LD area in Huh7 and HepG2 HCC cells²⁷⁴ and reduces lipid clearance from enterocytes.²⁷⁵ Recently, an *MTTP* variant c.1691T>C (rs745447480) encoding MTP Ile1564Thr was characterised in a family with hereditary NAFLD.²⁷⁶ Unlike what was previously discussed in the case of hepatitis C, the major allele (G) of *MTTP*-493G/T (rs1800591), associated with reduced transcription of *MTTP*, is linked to susceptibility to non-alcoholic steatohepatitis, but perhaps not steatosis alone, in dominant and heterozygote models of heredity.²⁷⁷ In addition, rs641738 downstream of the lysophosphatidylinositol acyltransferase 1 gene (*MBOAT7/LPIAT1*) is associated with hepatic steatosis and advanced fibrosis in NAFLD, although the data need to be validated in non-white populations,²⁷⁸ and is reportedly linked to liver injury in severe COVID-19.²⁷⁹ *LPIAT1* catalyses addition of archidonoyl CoA to lysophosphatidylinositol as part of the Lands cycle of phospholipids.²⁸⁰ The *GCKR*, *PPP1R3B*, *LYPLA1* and *NCAN* genes all have variants associated with hepatic steatosis on CT-based assessment.²⁸¹ *PNPLA3*, *MBOAT7* and *TM6SF2* mutations are also associated with alcohol-related cirrhosis, independent of the presence of type 2 diabetes, age, sex and BMI.²⁸²

In CHC, Stättermayer et al.²⁸³ found that homozygosity for a variant SNP, rs2645424, in a gene encoding an enzyme involved in cholesterol biosynthesis, farnesyl diphosphate farnesyl transferase 1 (*FDFT1*), was associated with advanced fibrosis only in CHC patients without hepatic steatosis, while homozygosity for *PNPLA3* rs738409 predicted presence of hepatic steatosis and, independent of hepatic steatosis, advanced fibrosis. The study was performed in patients with genotype 1 or genotype 4 HCV infection. This finding led Stättermayer et al. to suggest that metabolic involvement in CHC disease progression may differ according to the presence of hepatic steatosis. *FDFT1* rs2645424 had previously been identified as a variant associated with NAFLD activity score, a composite score dependent on extent of steatosis, inflammation and ballooning, in a cohort of white, non-Hispanic women recruited for suspected or confirmed NAFLD or cryptogenic cirrhosis.²⁸⁴

7 | THERAPEUTICS FOR MASLD

Treatment of MASLD is commenced with the intention of preventing liver-specific disease progression and managing risk of cardiovascular

disease. The first measure proven to be effective in MASLD is lifestyle change, including a healthy diet and regular exercise, with the goal of achieving and maintaining a healthy weight. If a patient is not able to achieve this with a prescribed and supervised programme, pharmacotherapy or bariatric surgery for eligible patients can be offered with discussion of risks and benefits. Management of other cardiovascular risk factors and comorbidities, such as type 2 diabetes, should be optimised.²⁸⁵ While HCC can develop in MASLD without cirrhosis, screening is reserved for people with MASLD and cirrhosis due to their high risk of HCC.^{64,285}

Importantly, diet-induced weight loss of 10% among obese patients with MASLD can reverse metabolic derangement by greatly reducing hepatic de novo lipogenesis, circulating insulin and intrahepatic triglycerides.⁴⁵ Non-obese patients with MASLD benefit from lifestyle changes and more moderate weight loss as well, related to the high frequency of visceral adiposity associated with abnormal waist circumference in these patients.^{285,343} Malnutrition related to diet or rapid weight loss following surgical intervention has been described in case reports as a rare cause of hepatic dysfunction in people with obesity.²⁸⁶ Historic use of jejunio-ileal bypass was associated with acute liver failure in 7% of patients,²⁸⁷ with development of hepatic steatosis following this procedure possibly attributable to the formation of a blind-ended bowel loop with bacterial colonisation.²⁸⁸ Prescribed exercise in a programme of at least 2 months in duration can reduce hepatic steatosis in patients with NAFLD,²⁸⁹ despite lower basal whole-body and hepatic fat oxidation and diminished response to acute exercise described in NAFLD subjects.²⁹⁰ Bariatric surgery may be helpful, particularly in obese patients who have MASH and fibrosis and who fail lifestyle modification, and in carefully selected patients with MASH cirrhosis.²⁹¹ According to a meta-analysis by Fakhry et al.,²⁹² receipt of bariatric surgery improves steatosis in over 80% of NAFLD patients and often improves steatohepatitis (38%–78% of patients) and fibrosis (21%–41% of patients). Limitations include the necessity of close monitoring in the post-surgical patient, ongoing vitamin and mineral supplementation and the observation that patients post-bariatric surgery may increase their alcohol consumption.²⁹¹

Treatment modalities for MASH may have pleiotropic effects on key drivers of progressive MASLD, including metabolic dysfunction, insulin resistance and inflammation. However, discussion of risks versus benefits of treatment are necessary. The PIVENS randomised controlled trial demonstrated that pioglitazone or vitamin E could reduce lobular inflammation and hepatic steatosis in people with non-alcoholic steatohepatitis (NASH) but without overt diabetes mellitus or cirrhosis and found significant improvement in a score of combined NASH histology features, but not fibrosis, after 96 weeks of 800 international units (IU) of vitamin E daily.²⁹³ Vitamin E functions as an antioxidant, while pioglitazone is a thiazolidinedione PPAR γ agonist that increases insulin sensitivity in the liver and periphery. However, long-term high-dose vitamin E therapy has been associated with increased mortality in the general population in a large meta-analysis^{294,295} and possibly enhances risks of prostate cancer²⁹⁶ and cerebral haemorrhage.²⁹⁷ Pioglitazone has adverse

effects including enhanced risk of bone fractures²⁹⁸ and potentially weight gain.²⁹³ Modern incretin-based treatments for diabetes may improve or stabilize histological parameters of MASH, such as the GLP-1 receptor agonist liraglutide, which was associated with resolution of NASH without worsening of fibrosis in a phase 2 study with limited enrolment.^{299,300} Results of a large randomised controlled trial of the GLP-1 receptor agonist semaglutide, the ESSENCE study, are forthcoming.^{300–302}

Future treatment modalities for MASLD may involve novel targeted endocrine therapies. A recent clinical trial has reported improvement in histological activity and fibrosis in people with NASH fibrosis without cirrhosis treated with a thyroid hormone receptor beta-selective agonist after 52 weeks of therapy,³⁰³ but full results of the trial after the interim analysis are not yet available. Similarly, treatment with a glycopegylated fibroblast growth factor 21 (FGF21) analogue has recently been reported to increase resolution of NASH and improve fibrosis after 24 weeks of therapy in patients with moderate or severe NASH fibrosis without cirrhosis.³⁰⁴ Serum levels of FGF21, which acts via the FGF receptor and co-receptor beta klotho to regulate glucose and lipid metabolism,³⁰⁵ are increased in people with metabolic syndrome.³⁰⁶ More data on clinical outcomes and adverse effects of longer-term treatment are needed to validate the results of the phase 2 randomised controlled trial.³⁰⁴

Free cholesterol in the liver and the free cholesterol-to-PC ratio increase progressively from MASLD to MASH,³⁰⁷ like the increases in hepatic free cholesterol that are seen in diabetes.³⁰⁸ Inhibition of HMG-CoA reductase with statins may be hepatoprotective,^{309–311} and their benefits in people with compensated liver disease outweigh the risk of rare hepatotoxicity associated with these drugs.^{312,313} Statins are primarily metabolised by the liver; further study is needed to determine safety and cardiovascular efficacy in people with cirrhosis and decreasing degrees of residual liver function.³¹⁴ Experimental inhibition of acetyl CoA carboxylase or fatty acid synthase has been described, but safety concerns related to hypertriglyceridaemia and platelet count decreases in the case of acetyl CoA carboxylase inhibitors, as well as adverse effects of chronically inhibiting *de novo* lipogenesis, such as xerophthalmia, dry skin and hair loss, have hampered drug development.³¹⁵

8 | FUTURE RESEARCH DIRECTIONS

Future studies will help to resolve important questions regarding the role of lipotoxicity in the pathogenesis of CLD, in which LD formation may play a protective role. Alternatively, increased cell stiffness associated with larger LDs^{316,317} may contribute to organ dysfunction. Fatty transformation may increase susceptibility to early graft dysfunction among recipients of liver grafts with a moderate-to-high degree of macrovesicular steatosis ($\geq 30\%$ liver occupancy).³¹⁸ It is speculated that large-droplet macrovesicular steatosis can cause hepatocytes to rupture on cooling and/or obstruct blood flow during donor liver reperfusion.³¹⁹ However, Choi et al.³²⁰ found that small-droplet macrovesicular steatosis, defined as one or more LDs that

each occupy less than half of the hepatocyte, in $\geq 30\%$ of liver hepatocytes independently correlated with risk of acute cellular rejection and chronic ductopenia, while moderate large-droplet steatosis involving up to 60% of liver hepatocytes did not. Neither measure was associated with changes in patient survival.³²⁰ Allografts with a degree of small-droplet macrovesicular steatosis greater than 15% additionally have less adenosine triphosphate content than those with less small-droplet steatosis, which is further reduced after reperfusion.³²¹

Importantly, accumulating literature suggests an association between cellular LD formation and exposure to environmental contaminants, such as diesel exhaust,³²² multi-walled carbon nanotubes,³²³ microplastics³²⁴ and ozone.³²⁵ SLD is both polyfactorial and highly prevalent, which can complicate identification of specific causal factors. Fatty transformation of hepatocytes could theoretically lead to increased susceptibility to agents that induce hypoxia and increased hepatic accumulation of lipid-soluble toxins, although preexisting liver disease does not generally increase the risk of a drug-induced liver injury⁴ (p. 812). A recent single-centre retrospective cohort study among U.S. paediatric patients with NAFLD identified associations between plasma concentrations of perfluoroalkyl substances, which form a class of persistent environmental contaminants, and likelihood of fibrosis and activity of steatohepatitis on liver biopsy, as well as alterations in plasma concentrations of products of glycerophospholipid and amino acid metabolism.³²⁶ No effect on liver steatosis was seen, although the evaluation of steatosis was semiquantitative.

9 | CONCLUSIONS

In summary, hepatic steatosis is associated with a variety of acute and CLDs. The presence and number of steatogenic risk factors, including genetic background, may predispose to multiple causes of liver disease. Correlation with available clinical, familial and medication histories is recommended when diagnosing SLD to identify possible causes and guide further diagnostic testing. Diet and exercise, which increase insulin sensitivity and reduce adiposity, are essential modifiable behaviours that can reverse chronic hepatic steatosis. Greater accessibility of non-invasive assessment of hepatic steatosis will help clinicians and researchers determine links among hepatic steatosis, other aspects of the metabolic syndrome and clinical outcomes.

A limitation of this review is the lack of a systematic search strategy when reviewing the literature. The reader is referred to recent research reviews³²⁷ published on important aspects of LD biology that are not fully addressed in this review, including LD interactions with mitochondria,³²⁸ LD function in polyunsaturated fatty acid trafficking,^{329,330} and the role of LDs in host-pathogen dynamics.³³¹ Potential therapeutics for MASLD are the ongoing subject of research and are reviewed elsewhere,³³² as are candidate therapies for alcohol-related hepatitis³³³ and ALD.³³⁴ The criteria for diagnosing MASH are outside the scope of this review and are based on the

pioneering work of Brunt et al.³³⁵ The management of acute liver failure of known and unknown aetiologies is a complex topic that has recently been reviewed by Tujios, Stravitz and Lee.³³⁶

Improved understanding of the cellular biology of LDs will elucidate mechanisms of the aetiological agents of acute and chronic liver failure associated with steatosis and expose therapeutic targets. The molecular mechanisms of disruption of cellular lipid homeostasis and LD dynamics in SLD remain to be characterised, as are the contributory mechanisms of genetic variants like those found within *PNPLA3*. The protein and lipid composition of LDs in microvesicular versus small-droplet and large-droplet macrovesicular steatosis and their relationship to the adipose tissue and plasma lipidomes are of great interest but require special sample preparation. Alternatively, the non-invasive assessment of hepatic steatosis by magnetic resonance spectroscopy can allow for detailed metabolomic study and characterisation of hepatic lipid composition.³³⁷ Investigation of peroxisomal beta oxidation, which is a significant contributor to metabolism of very-long-chain fatty acids,³³⁸ could also be important. Interestingly, seipin and its interactor, PEX30, may be involved in the biogenesis of both pre-peroxisomal vesicles and LDs from the ER,^{339,340} and proteins involved in peroxisome biogenesis have also been implicated in segregating LD-associated proteins into specific ER subdomains.¹¹⁶

While this article was in press, Dhiman et al.¹¹⁷ published a series of elegant biochemical and modeling experiments demonstrating that the monotopic protein UBXD8 adopts different conformations in its hydrophobic hairpin domain depending on its membrane environment, with the antiparallel alpha helices adopting a more acute angle to each other when embedded in a phospholipid bilayer than in monolayers enclosing a neutral lipid core. They suggest that the transition between these two states poses a high energetic barrier that may necessitate participation of seipin in sorting and/or properly conforming UBXD8 as it moves from the ER membrane onto nascent LDs. They further speculate that the cage-like initial structure of the seipin oligomer (Figure 3B) might block other LD-associated proteins incapable of the deep-to-shallow conformational change from sorting to LDs early in LD biogenesis. The results of their study and others presented in this review suggest that LDs are complex and evolving organelle-like structures incorporating lipids and select membrane proteins, requiring active processes to regulate their biogenesis, subcellular interactions and fate, which may be important in preventing lipotoxicity and the progression of SLD.

AUTHOR CONTRIBUTIONS

ABK wrote the paper and designed the figures, ABK, AT, HA, FM, ZMY and ZDG provided manuscript revisions and assisted in topic selection and ZMY and ZDG provided supervision and expert opinion. ABK takes responsibility for the integrity of the contents of this manuscript and the process of producing it. All authors approved the final version of the manuscript including the authorship list.

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